Too Fast to Read:
Topic Choice and the Incidence of Research Tools

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Abstract

Reading has two jobs in science. It *teaches*, raising the quality of new work, and it *protects*, revealing projects that others have already done. We model researchers who choose a topic, split their time between producing and reading, and must read a topic's literature before they can join it. For any learning and detection technology with bounded absorption, the payoff of working in a topic is single peaked in the flow of its literature, and free choice of topics has the structure of a system of cities. Two results follow. First, a speed limit: no newcomer can catch up with a literature that grows faster than she reads, so such topics close, rents near the limit grow without bound, fast-growing topics overshoot and slow ones never take off. Second, frontier incidence: output per researcher equals the value of opening a new topic, so a research tool is worth what it adds to that value. Inside a topic every speed-up is worth the same per hour saved; entry ranks them. Production tools survive almost fully; reading tools survive mainly through founders who stay in the topics they open, and catch-up tools least. Valued by readership rather than credit, the ranking reverses. On arXiv, more volume opens new ground rather than thickening old topics. On 2.2 million submissions with exact timestamps, four arXiv rule changes and design features identify the margins the model relies on. A tighter endorsement rule cut first-time submitters by about a sixth. A peer-review requirement halved review articles in computer science. A cap of two submissions per person per month binds on about one paper in a hundred. Daily listing position barely moves readership and leaves the detection of overlapping work unchanged. In the calibrated model, per hour saved, a production tool is worth 1.3–7.2 times a reading tool, and the static gain from optimal topic choice is small.

JEL: O31, O33, O40, D62, I23, R12. **Keywords:** economics of science, topic choice, duplication, system of cities, burden of knowledge, incidence, artificial intelligence, arXiv.

1 Introduction

Submissions to arXiv more than doubled between 2018 and 2025, and in September 2026 the archive received a record 40,363 papers. Large language models now produce research text faster than any researcher can, and they are starting to read and summarise for us too. Two opposite hopes and fears follow. Faster production might flood science with duplicates. Better reading tools might be the remedy and raise research productivity. This paper argues that free choice of research topics decides between them, and that what matters about a tool is not how much time it saves but whom it helps: the people who open new topics, or the people who join existing ones.

The argument rests on one observation: researchers choose not only how much to read and produce but *where* to work. Reading has two jobs. It *teaches*: what one reads raises the quality of one's own work, and more so in a topic with more to learn from. It *protects*: each earlier paper may already cover one's project, and reading is how one finds out. Unaware duplicates lose part of their credit; detected overlaps can be turned into follow-ons that cite the earlier work, or abandoned. These two jobs make the payoff of working in a topic rise with the flow of its literature while there is much to learn, and fall once more of that flow goes unread. Newcomers must also read a topic's accumulated literature before they can contribute, and they enter topics until they are no better off than at the frontier, where new topics are opened by founders who then stay in them.

The results. Three results organise the paper.

Frontier incidence. In a stationary equilibrium every option a researcher takes is worth the same, so output per researcher equals the value of opening a new topic (Lemma 4). A research tool is therefore worth, in the long run, what it adds to that value (Theorem 3). Inside a topic, every speed-up is worth exactly the time it saves (Lemma 5): because researchers spend little of their time reading, a faster reader gains little, but per hour saved she gains as much as a faster producer. Entry then ranks the tools. A tool that speeds up production raises the value of everything a researcher does, including founding a topic, and survives entry almost entirely. A tool that speeds up reading reaches the frontier only through founders, who read in the topics they open. A tool that speeds up catching up on a literature reaches it least, since founders never have to catch up. The share of a reading gain that survives entry is at least the volume elasticity of density, which the data measure, and rises with the share of the frontier's value that founders earn as incumbents.

A speed limit. A newcomer can catch up with a literature only if it grows more slowly than she can read it (Lemma 2). Topics whose flow exceeds that speed are closed to newcomers, however rewarding they are, and incumbents' rents near the limit grow without bound (Theorem 2). A tool that lets everyone produce faster raises every topic's flow at once and can push the densest topics past the limit. Because the stock of literature lags its flow, topics that grow fast overshoot their best density, even when newcomers see it coming, and topics that grow too slowly stall before they are worth joining and are abandoned (Proposition 9).

Units. Measured by credited output, faster production scales science. Measured by readership, it adds nothing, because total reading is fixed by researchers' time and more papers share it; then

only reading tools raise output (Proposition 8). We report both.

What is new, and what is not. The equilibrium structure is that of a system of cities (Henderson, 1974) or of clubs (Buchanan, 1965). The average payoff is single peaked in group size, new groups enter freely, equilibria are stable only on the falling side, the planner sits at the peak of the average payoff, and an entry tax decentralises the optimum. Self-organised cities are too large for the same reason (Henderson and Becker, 2000; Scotchmer, 2002). We use this structure and claim four things that it does not provide. First, a microfoundation of the hump in the two jobs of reading, valid for any learning and detection technology in a broad class. The proof uses joint log-concavity of the researcher's problem in reading time and density (Theorem 1), and the hump comes from bounded absorption against a covering hazard that grows with the literature. Second, the crowders are the readers: the sign of the spillover from one more paper is derived from the envelope theorem rather than assumed, and it changes along the payoff curve (Proposition 1). Third, the duplication rate is pinned by entry, with an exact volume elasticity of density built from observables (Proposition 3). Fourth, the frontier-incidence theorem and the speed limit, which together say which research tools scale science and when a literature grows too fast to join. The catch-up cost, the need to read a topic's stock before contributing (Jones, 2009), adds a wedge that the city literature lacks: incumbents' returns rise with density even where newcomers find topics too crowded (Corollary 1).

Evidence. We confront the model with the arXiv record in two ways. The descriptive evidence (Section 7) comes from 1.49 million papers posted in 2018–2025, classified into 150 text-based areas. The anchor fact is that volume opens new ground. From the 95th to the 25th percentile similarity radius, local density rises with area volume at an elasticity of 0.12–0.19, against 0.53–0.81 if papers kept landing in the same idea space. At the closest-overlap radius the estimate is 0.035, though it is imprecise (SE 0.118). Returns are hump-shaped in topic density. Unaware duplicates lose about a fifth of their citations, close to the 21% penalty for scooped teams in Hill and Stein (2025a), while aware follow-ons gain. Once corrected for citation recall, about 60% of close overlap is aware.

The quasi-experimental evidence (Section 8) uses arXiv's own rules on a new panel of 2.2 million arXiv submissions with exact timestamps through 15 October 2026. Four shocks speak to the margins the model relies on.

- *Entry cost.* In January 2026, arXiv stopped endorsing new submitters on the strength of an institutional e-mail address alone; Mathematics had moved in December 2025. The share of papers by first-time submitters fell by 3.4 percentage points (about 15%) in the affected sections, net of the year before. The staggered timing matches the rule.
- *Catch-up cost.* From November 2025, review and position papers in computer science needed prior peer review. Review-type papers in computer science halved relative to other archives, more so in categories that had relied on them. First-time authorship fell more in those categories, though a smaller pre-trend of the same sign makes this suggestive.

- *Writing cap.* From October 2026, each person may submit at most two papers a month. The cap removed about 1.3% of papers. Total volume kept growing at its usual seasonal pace, so per-submitter caps are a weak lever on volume.
- *Readership.* A paper's position in the daily listing depends on how many others submitted earlier that day. We use this to test the detection channel directly. Position has only a small effect on citations, and no detectable effect on later overlapping work or on the share of it that cites the paper, with tight bounds. Overlap is detected by targeted search rather than by browsing.

At the author level, adopters of LLM-style writing raised their output by about 7% relative to non-adopters. That is far less than the 36% of Kusumegi et al. (2025), whose design has been criticised for mechanically generating positive dynamics (Renault et al., 2026).

Magnitudes. The calibrated model is disciplined by the corrected arXiv moments and by survey evidence that researchers spend about 7% of their working time reading articles (Tenopir and King, 2008) (Section 9). At a given topic, a 30% faster producer gains 26.2 log points and a 30% faster reader 1.96: the same per hour saved. In the long run the producer keeps almost all of it. The reader keeps 14% of her gain if explorers never stay in the topics they open, and 74% if they stay with probability one half; catch-up tools keep 40%. Per hour saved, a production tool is worth between 1.3 and 7.2 times a reading tool. On the simulated life cycle, where founders spend much of their tenure in topics still growing at a pace set by diffusion, catch-up tools barely reach the frontier, and the rents of diffusion-limited entry let more of every gain survive. Valued by readership instead, the ranking reverses: a 30% faster pen adds 1.3 log points and 30% faster reading 21.2. The static welfare cost of crowding is small: optimal topic choice would raise output per researcher by 1.2%, and overlap destroys about 2.9% of credit at mature density. Mature topics sit at 17% of the speed limit, the most fertile at 33%. With the measured volume elasticity, topic crowding explains little of the decline in research productivity.

Implications. Section 10 collects three further results. Topics that grow faster than a threshold rate overshoot their best density, and rational entrants overshoot more than myopic ones once a topic grows fast, because what deters entry is the stock, and a young topic has little of it. The stepping-on-toes parameter of growth theory has two measurable versions, depending on whether open topics count as knowledge, and both are close to one. And policy has to depend on a topic's age. A flat subsidy to the frontier, optimal in a stationary economy, is almost as large as the margin by which a newborn topic beats the frontier in the calibration; it raises the bar for the early followers who carry young topics to maturity, and so kills more of them. A tax on joining topics in proportion to their stock is zero at birth and reaches most of the optimum without that side effect (Proposition 11).

Related literature. The paper joins five literatures. The first is the economics of science and priority (Merton, 1957, 1961; Dasgupta and David, 1994; Stephan, 1996; Hill and Stein, 2025a,b; Bobtcheff et al., 2017) and the philosophy of the division of cognitive labour (Kitcher, 1990; Strevens, 2003). We add reading as a choice with two distinct roles, and derive rather than assume the costs of duplication. Hill and Stein (2025b) show that priority races lower quality; our cost of crowding works through detection and catch-up instead, and the two are complementary. Chu and Evans (2021) document that progress slows in large fields as attention stays with established work, which our speed limit rationalises. The second is the theory of excess entry, common pools, cities and clubs (Gordon, 1954; Weitzman, 1974; Mankiw and Whinston, 1986; Henderson, 1974; Buchanan, 1965; Henderson and Becker, 2000; Scotchmer, 2002) and of patent races (Loury, 1979; Dasgupta and Stiglitz, 1980). Our resource is attention to a literature, and the crowders are its readers; spatial models of endogenous agglomeration (Lucas and Rossi-Hansberg, 2002; Allen and Arkolakis, 2014) suggest how topics could form in a continuous idea space. The third is the direction of research (Bryan and Lemus, 2017; Hopenhayn and Squintani, 2021; Carnehl and Schneider, 2025; Bramoullé and Saint-Paul, 2010; Rzhetsky et al., 2015; Foster et al., 2015; Myers, 2020; Borjas and Doran, 2012, 2015; Azoulay et al., 2019). Carnehl and Schneider (2025) model the choice of question by novelty and feasibility. We model the choice by density, derive the hump, and measure the margin. Borjas and Doran (2015) document that researchers move in idea space in response to crowding, which is the margin our model turns on. The fourth is the macroeconomics of ideas and of access to knowledge (Romer, 1990; Aghion and Howitt, 1992; Jones, 1995; Kortum, 1997; Weitzman, 1998; Bloom et al., 2020; Jones, 2009; Ekerdt and Wu, 2024; Furman and Stern, 2011; Murray et al., 2016; Iaria et al., 2018; Biasi and Moser, 2021; Evans, 2008; McCabe and Snyder, 2015; Agrawal et al., 2016; Park et al., 2023). We give stepping on toes an equilibrium foundation and explain when better access to the literature raises productivity: when it reaches the people who open new topics. The fifth is artificial intelligence in science (Agrawal et al., 2019; Brynjolfsson et al., 2019; Jones, 2025; Duede et al., 2026; Hao et al., 2026; Liang et al., 2024; Kusumegi et al., 2025). Jones (2025) studies AI in a task-based R&D production function, and Duede et al. (2026) model LLMs as labour-augmenting technology that shifts effort across projects. We separate production, entry, reading, search and catch-up channels by where their gains end up in equilibrium. Methodologically, our text-based density measures follow Hoberg and Phillips (2016) and Kelly et al. (2021), and the listing-position design builds on Haque and Ginsparg (2009, 2010) and Feenberg et al. (2017).

Section 2 sets up the model and Section 3 derives the payoff curve. Sections 4 and 5 characterise equilibrium, the speed limit, welfare and social value. Section 6 states the frontier-incidence theorem. Sections 7 and 8 present the descriptive and quasi-experimental evidence, Section 9 quantifies the model, Section 10 collects the implications for topic life cycles, growth and policy, Section 11 lists out-of-sample predictions, and Section 12 concludes.

2 Model

The economy is stationary and time is continuous. Within a topic the relevant literature is a flow of papers, so we describe a researcher's choices per unit of time.

2.1 Researchers, topics and time

There is a continuum of topics. Topics that are open are ex ante identical. A mass N of researchers each has one unit of time per period, split into a share x spent producing research and a share $r = 1 - x$ spent reading. Producing full time yields a projects per period. The parameter a is research output per unit of production time: doing the research and writing it up. It is not drafting speed. A tool that drafts text faster raises a in proportion to the share of production time that goes into drafting. Topic j hosts n_j researchers who produce, and receives a flow of

$$P_j = a\, x\, n_j \tag{1}$$

papers per period. P_j is the topic's *density*. We call the ratio of production speed to reading speed, a/b, the *pace* of science.

Density is a flow. What a researcher learns from, and what can cover her project, is her topic's current literature: recent papers, which carry its current methods and results. What a newcomer must master before she can contribute is the topic's *stock*, its accumulated relevant literature (Section 2.5). The distinction is the one between keeping up with a field and getting into it. In a stationary topic the stock is proportional to the flow, so the distinction matters only for topics that grow or shrink. Appendix B.4 lets learning draw on the stock as well.

2.2 Reading teaches

A researcher who reads full time reads b papers per period, so a reading share r yields $R = b\, r$ papers drawn from her topic's flow. What she reads raises the quality of her own work by the factor $Q(R, P)$.

Assumption 1 (Learning)**.** $Q : \mathbb{R}^2_+ \to [1, \infty)$ is continuous, twice continuously differentiable on $\mathbb{R}^2_{++}$, nondecreasing in both arguments, with $Q(0, P) = Q(R, 0) = 1$ and $\bar{Q}(R) \equiv \sup_P Q(R, P) < \infty$ for every R. Moreover $\ln Q$ is jointly concave and supermodular on $\mathbb{R}^2_+$.

Nothing is learned without reading or without a literature to read, and what one can learn is bounded by what one reads. Concavity of $\ln Q$ rules out increasing returns to reading. Supermodularity says that reading is worth more in a topic with more to learn from. Our running example is

$$Q(R, P) = \left(1 + \beta\, h(R, P)\right)^{\gamma}, \qquad h(R, P) = \frac{RP}{R + P}, \qquad \beta > 0,\ \gamma \in (0, 1]. \tag{2}$$

The harmonic aggregator h is close to R when the reader samples a small part of a rich literature and close to P when she reads a thin literature exhaustively. Because h is concave and homogeneous

of degree one, the example satisfies Assumption 1 (Appendix A).

2.3 Reading protects

Each earlier paper in the flow *covers* a given project independently with a small probability, so the number of earlier covering papers is Poisson with mean θP. A covering paper is caught only if the researcher finds it. With reading coverage $c = R/P$ it is caught with probability $\varphi(c)$.

Assumption 2 (Detection)**.** $\varphi : \mathbb{R}_+ \to [0,1]$ is continuous and twice differentiable on $(0,\infty)$, with $\varphi(0) = 0$, $\varphi' > 0$ and $\varphi'' < 0$.

Concavity says that the first papers one screens are the most informative about overlap. Random reading with replacement gives $\varphi(c) = 1-e^{-c}$. Our example $\varphi(c) = 1-e^{-\sigma c}$ indexes *search precision* by σ: a better search engine finds covering papers sooner. Our evidence (Section 8) is that overlap is found by deliberate search rather than by browsing new announcements. We therefore read R as the part of the literature a researcher screens, mostly by targeted search, and φ as the result of that search. Appendix B.5 treats search that must be repeated for every project, in which case production speed no longer drops out of the payoff curve.

Some overlap cannot be prevented by reading. A competing project that is in progress elsewhere, and appears only after one's own project has started, cannot be found in the literature. Such *concurrent* covering projects arrive at rate $\theta_c P$.

Credit. Overlap erodes credit multiplicatively. Each earlier covering paper that goes undetected turns part of the project into a duplicate and removes a share λ of its credit. Each concurrent one removes a share λ_c. When an earlier covering paper is caught, the researcher chooses between two options. She can *pivot* to a fresh variant at a time cost that leaves a share $1-\kappa_p$. Or she can *proceed* as a follow-on that cites the earlier paper and keeps a share δ drawn from a distribution G. She proceeds if $\delta > 1-\kappa_p$. The expected share kept after a detection is $1-\ell \equiv \mathbb{E}\max\{\delta, 1-\kappa_p\}$.

Assumption 3 (Credit)**.** $0 \le \ell < \lambda \le 1$, $\theta > 0$, $0 \le \lambda_c \le 1$ and $\theta_c \ge 0$.

Detection is valuable ($\ell < \lambda$), and duplicates need not earn zero ($\lambda < 1$ is allowed). The arXiv evidence (Fact 3) requires both. Credit here is private. Section 5 separates it from social value, since part of the credit lost to overlap may be a transfer to the earlier paper. Let $\Delta \equiv \lambda - \ell$ and $q \equiv 1 - G(1-\kappa_p)$, the share of detections that proceed.

Lemma 1 (Credit and overlap)**.** *Under Assumptions 2 and 3, expected credit per project is*

$$C(R,P) = \exp\Big\{ -\theta P\big[\lambda - \Delta\,\varphi(R/P)\big] - \theta_c \lambda_c P \Big\}. \tag{3}$$

*Earlier covering papers that the researcher does not pivot away from (*flagged overlap*) arrive at rate* $\theta P\,[1-(1-q)\varphi]$*. A share*

$$A = \frac{q\,\varphi}{1-(1-q)\,\varphi} \tag{4}$$

of them is aware*: the later paper cites the earlier one. Unaware duplicates keep a share* $1-\lambda$ *of credit. Aware follow-ons keep* $\mathbb{E}[\delta \mid \delta > 1-\kappa_p]$*, which can exceed one.*

The lemma follows from Poisson thinning. Detected and undetected covering papers are independent Poisson counts with means $\theta P\varphi$ and $\theta P(1-\varphi)$, and $\mathbb{E}[t^K] = e^{-\mu(1-t)}$ for $K \sim \text{Poisson}(\mu)$. Only $\Theta \equiv \theta\lambda$, Δ/λ and $\Theta_c \equiv \theta_c\lambda_c$ enter payoffs. The arXiv penalty for unaware duplicates of earlier work (Fact 3) identifies λ. The 21% penalty that Hill and Stein (2025a) estimate for scooped structural-biology teams is a penalty for losing a race, so it speaks to λ_c, not to λ. The two numbers are close, but they measure different events.

The detection machinery does three jobs in what follows. It maps the observable aware share and flagged-overlap rate into density, which is what makes the invariance of overlap in Proposition 3 a testable restriction and what disciplines σ in the calibration. It separates search tools, which raise the aware share at given density, from other reading tools (Proposition 7). And it supplies the falling arm of the payoff curve (Theorem 1). The incidence results of Section 6 do not depend on it: they hold for any payoff curve.

2.4 The researcher's problem

A researcher in a topic of density P earns credit $a\,x\,Q(R,P)\,C(R,P)$ per period: producing x projects is worth x, and each is worth QC. Dividing by the common factor a,

$$g(r,P) \equiv (1-r)\,Q(br,P)\,C(br,P), \qquad v(P) \equiv \max_{r\in[0,1)} g(r,P). \tag{5}$$

The *payoff curve* v is the central object of the paper. Production speed a scales every payoff and so drops out of v. The researcher takes P as given: she ignores both that her papers teach her colleagues and that they add to what her colleagues must read.

Remark 1 (Topic size and what density measures). A topic has a fixed set of projects: the coverage rate θ is a property of the topic and does not shrink when the topic attracts more people. This is a normalisation, not a restriction, once density is measured locally. Let a field of breadth B consist of B cells of unit breadth, each receiving a flow P/B, and let researchers read and be covered within their own cell, so that the coverage rate per paper is θ_0/B. The field is then B independent copies of a unit topic with density P/B, and every result below holds with P replaced by P/B. The model's topic is therefore a neighbourhood of fixed size in idea space, and P is papers per unit of idea space: the quantity that nearest-neighbour similarity measures. If papers fall at rate P/B in a d-dimensional space, the distance to the k-th nearest earlier paper scales as $(kB/P)^{1/d}$. "Opening a topic" means moving into an empty cell, and the question of whether topics broaden as they grow becomes the question of how the volume elasticity varies with the similarity radius, which Section 7 measures and Appendix B.3 derives.

2.5 Careers, catch-up and the speed limit

Careers in a topic end at rate ρ (retirement, a change of field), and topics become obsolete at rate ξ, which releases their members. The literature a newcomer must master is a stock S that accumulates the flow and loses relevance at rate δ: $\dot{S} = P - \delta S$, so $S = P/\delta$ in a stationary topic. She reads it at speed b_c before she can contribute. This is the *catch-up* cost, a topic-level version of the burden of knowledge of Jones (2009). Papers that appear while she reads must be read as well, so her backlog follows

$$\dot{B} = P - \delta B - b_c, \qquad B(0) = S, \tag{6}$$

and catch-up ends when the backlog is cleared.

Lemma 2 (Catch-up and the speed limit)**.** *With a constant flow P and an initial stock $S > 0$, catch-up takes*

$$T(P,S) = \frac{1}{\delta}\ln\Big(1 + \frac{\delta S}{b_c - P}\Big) \quad \text{if } P < b_c, \qquad T(P,S) = \infty \quad \text{if } P \geq b_c. \tag{7}$$

In a stationary topic $T(P) = -\delta^{-1}\ln(1 - P/b_c)$. A newcomer who joins a stationary topic of density P and then earns $a\,v(P)$ per period obtains, per period of her spell in the topic, the expected payoff $a\,w(P)$, where

$$w(P) \equiv v(P)\,\Gamma(P), \qquad \Gamma(P) \equiv e^{-(\rho+\xi)T(P)} = \Big(1 - \frac{P}{b_c}\Big)^{\nu}, \qquad \nu \equiv \frac{\rho+\xi}{\delta}. \tag{8}$$

A share $(1 - P/b_c)^{\rho/\delta}$ of a stationary topic's members have caught up and produce, so the topic has $n(P) = P\,(1 - P/b_c)^{-\rho/\delta}/(a\,x^*(P))$ members, where $x^*(P)$ is the optimal production share.

Catch-up is a race between the reader and the literature. To first order in P/b_c the catch-up time is $P/(\delta b_c)$, the stock divided by the reading speed. That approximation ignores the papers that arrive during catch-up and understates T, the more so the faster the topic. The exact version has a *speed limit*: no newcomer ever catches up with a topic whose literature grows faster than she can read it. Theorem 2 draws out its consequences. The frictionless benchmark is $b_c \to \infty$, where $\Gamma \equiv 1$. Incumbents earn $a\,v(P)$ per period; newcomers earn $a\,w(P)$ on average over their spells. Catch-up reading earns admission and nothing else: it is not counted as learning. Appendix B.4 relaxes this.

2.6 The frontier

Instead of joining an open topic, a researcher may explore. An explorer produces e per unit of production speed per period: exploratory work that does not found a lasting topic. She founds a viable topic at rate s. A *founder stays* in the topic she opens. She wrote its first papers, so she needs no catch-up, and she remains an incumbent until her career ends or the topic dies. Topics are born at rate sE, where E is the measure of explorers, and die at rate ξ. In a stationary state

births equal deaths:

$$s\,E = \xi\,M, \tag{9}$$

where M is the measure of open topics. The frontier is a flow, like the topics it feeds.

Lemma 3 (The value of the frontier)**.** *The value of exploring, per unit of production speed and per period of the spell it starts, is*

$$F = (1-\omega)\,e + \omega\,\bar{U}_F, \qquad \omega \equiv \frac{s}{\rho+\xi+s}, \tag{10}$$

where $\bar{U}_F = (\rho+\xi)\,\mathbb{E}\int_0^\infty e^{-(\rho+\xi)t} v(P_F(t))\,\mathrm{d}t$ *is the founder's average payoff along the path* $P_F(t)$ *of the topic she opens.*

The share of the frontier's value that founders earn as incumbents,

$$\theta_F \equiv \frac{\omega\,\bar{U}_F}{F} \in [0,1), \tag{11}$$

is the statistic on which the incidence of research tools turns (Section 6). A model in which pioneers never read and never stay in their topics has $\theta_F = 0$. As more directions are taken, a fruitful new one is harder to find (Kortum, 1997; Bloom et al., 2020). We let the frontier's value fall with the number of open topics at elasticity $\eta_F \geq 0$, so that $F(M,P) = (M/\bar{M})^{-\eta_F}\left[(1-\omega)e + \omega\bar{U}_F(P)\right]$, where the dependence on P comes from the topics founders end up in. In the stationary economy a newborn topic reaches the common density P at once, so $\bar{U}_F = v(P)$. Section 10 and the quantification put founders on the simulated life cycle of their topics.

Lemma 4 (Output equals the frontier)**.** *In a stationary equilibrium in which every option a free researcher takes is worth* F*, output per researcher is exactly* $a\,F$*:* $Y = a\,N\,F$*.*

The lemma is the renewal-reward theorem. Each researcher moves through spells: exploring and possibly founding, or catching up and producing in an open topic. The value of an option is the ratio of its expected output to its expected length, so a researcher's long-run output per period is the common value F of the options she takes. In a stationary population the cross-section average equals the time average. Lemma 4 turns every question about long-run productivity into a question about the frontier.

2.7 Equilibrium

Definition 1 (Topic equilibrium)**.** A stationary topic equilibrium is a density P^* for every open topic, a production share x^*, a measure M^* of open topics and a measure E^* of explorers such that: (i) $x^* = x^*(P^*)$ solves (5); (ii) newcomers are indifferent between joining any open topic and exploring: $w(P^*) = F(M^*, P^*)$; (iii) the allocation is stable under entry: $w'(P^*) < 0$; (iv) births equal deaths, $sE^* = \xi M^*$, and researchers add up: $E^* + M^* n(P^*) = N$.

Table 1: Notation

a, b	projects produced / papers read per period at full time	P	topic density (flow of papers per period)
x, r	production and reading shares, $r = 1 - x$	$R = br$, $c = R/P$	papers read; reading coverage
$Q(R, P)$	quality gain from what is read	$\varphi(c)$	probability an earlier covering paper is caught
θ, θ_c	coverage rates of earlier and concurrent work	λ, ℓ, λ_c	credit lost per unread / caught / concurrent covering paper
$g(r, P)$, $v(P)$	payoff per unit of a; its maximum	$\hat{P}$, $\hat{v}$	peak of v and its height
ρ, ξ	end of careers in a topic; obsolescence of topics	S, δ	stock of literature; decay of its relevance
b_c, T	catch-up reading speed (speed limit); catch-up time	Γ, ν	newcomer's survival factor $(1-P/b_c)^\nu$; $\nu = (\rho + \xi)/\delta$
$w(P)$	newcomer's payoff, $v(P)\Gamma(P)$	$\hat{P}_w$, $\hat{w}$	peak of w and its height
e, s	explorers' output; founding rate	ω, $\bar{U}_F$	founders' probability; founders' payoff
F, θ_F	frontier value; founders' share of it	η_F	depletion of the frontier in topics opened M
P^*, P^m	equilibrium (mature) density	$\mathcal{E}$	elasticity of density with respect to N
g, g_0, g^*	topic diffusion rate; take-off and overshooting thresholds	λ_J	stepping-on-toes parameter of Jones (1995)
λ_s, τ	social loss per unread covering paper; transfer share	α, $\mathcal{A}$	weight of readership in value; reads per paper

Condition (iii) selects allocations that survive the adjustment $\dot{n}_j \propto w(P_j) - F$ topic by topic. A single topic is too small to move F, and at a crossing where w rises one more entrant raises the payoff of entering and draws in more. Topics in equilibrium need not have the same density a priori, but (ii) and (iii) leave a single admissible density, so they do. We state comparative statics in the *large-topic limit*, in which explorers per topic, ξ/s, are small relative to members per topic, so that $M^* \approx N/n(P^*)$. Appendix A gives the general expressions.

3 The payoff curve

Every result in the paper rests on the shape of v. The next theorem shows that it is single peaked for every learning and detection technology that satisfies Assumptions 1 to 3.

Assumption 4 (Interior reading and learning potential)**.** (a) Reading the first paper is worth its time: $b\,[\partial_R \ln Q(0, P) + \theta\Delta\varphi'(0)] > 1$ for all $P > 0$. (b) A thin literature is worth reading: for some $\varrho > 0$,

$$\nu_0(\varrho) \equiv \lim_{P\downarrow 0} \frac{Q(\varrho P, P) - 1}{P} - \frac{\varrho}{b} - \theta\big[\lambda - \Delta\varphi(\varrho)\big] - \theta_c\lambda_c > 0.$$

Part (b) compares, at the first papers of a topic, the gain from learning along the ray $R = \varrho P$ with the time spent reading and the overlap risk. In the example it reads $\sup_\varrho\{\gamma\beta\varrho/(1+\varrho) - \varrho/b - \theta[\lambda - \Delta(1 - e^{-\sigma\varrho})]\} > \theta_c\lambda_c$.

Theorem 1 (The payoff curve)**.** *Under Assumptions 1 to 3:*

(i) v is continuous on $\mathbb{R}_+$, $v(0) = 1$, $v(P) \geq e^{-(\theta\lambda + \theta_c\lambda_c)P} > 0$ and $\lim_{P\to\infty} v(P) = 0$.

(ii) $\ln v$ *is concave on* $(0,\infty)$*. Under Assumption 4(a) it is strictly concave.*

(iii) Under Assumption 4, there is a unique $\hat{P} \in (0,\infty)$ *such that* v *is strictly increasing on* $[0,\hat{P}]$ *and strictly decreasing on* $[\hat{P},\infty)$*, with* $\hat{v} \equiv v(\hat{P}) > 1$*. For every* $F \in (0,\hat{v})$ *the equation* $v(P) = F$ *has exactly one root above* $\hat{P}$*.*

(iv) (Envelope) v *is differentiable on* $(0,\infty)$ *with*

$$\frac{v'(P)}{v(P)} = \underbrace{\partial_P \ln Q(R^*,P)}_{\text{more to learn from}} - \underbrace{\theta\big[\lambda - \Delta(\varphi(c^*) - c^*\varphi'(c^*))\big] - \theta_c\lambda_c}_{\text{more to miss}}, \tag{12}$$

where $R^* = b\,r^*(P)$ *and* $c^* = R^*/P$*.*

(v) The optimal reading share $r^*(P)$ *is unique and nondecreasing in* P*. Papers per researcher,* $a\,x^*(P)$*, are nonincreasing in density.*

The proof (Appendix A) rests on one observation. $\ln g$ is *jointly* concave in reading time and density. $\ln(1-r)$ and $\ln Q$ are concave by assumption. The erosion exponent $-\theta\lambda P + \theta\Delta\, P\varphi(R/P) - \theta_c\lambda_c P$ is linear plus the perspective of the concave function φ, and perspectives preserve concavity. Maximising a jointly concave function over one argument leaves a concave function of the other, so $\ln v$ is concave. A log-concave function that starts at one, rises, and vanishes at infinity is single peaked. Part (v) follows from Topkis's theorem: $\partial^2 \ln C/\partial R\,\partial P = -\theta\Delta\varphi''(c)\,c/P \geq 0$ and $\ln Q$ is supermodular, so reading and density are complements.

The theorem needs no parametric assumption, and its conditions are economic: no increasing returns to reading, and diminishing returns to search. Note what it does *not* need. Duplicates may keep most of their credit, since only $\lambda > \ell$ is required. The single peak comes from overlap that goes unread, not from an assumption that overlap is worthless.

Remark 2 (Where the hump comes from, and what the theorem rules out)*.* The fall in v comes from *bounded absorption.* What a researcher can learn is bounded in density ($\bar{Q}(R) < \infty$) and what she can read is bounded by b, while the expected number of covering papers, θP, grows linearly with the literature. Any model with these two features has a falling payoff at high density. Joint concavity of $\ln Q$ is what makes the rise and the fall meet at a single peak. It rules out complementarities among the papers one reads, for example a literature that becomes useful only once a critical mass of results can be combined. With such complementarities $\ln v$ can have several local peaks, topics can have more than one stable size, and the comparative statics below apply locally around each falling-side crossing. We keep the concave case. It is the natural benchmark, and the arXiv returns profile (Fact 2) shows a single hump.

The value of time. The envelope theorem also prices every speed-up inside a topic.

Lemma 5 (Time shares)**.** *At a fixed density,* $\partial \ln v/\partial \ln b = r^*(P)/(1-r^*(P))$*, while a faster production speed raises a researcher's payoff by* $\mathrm{d}\ln a$*. A speed-up of a task that takes a share* s_k

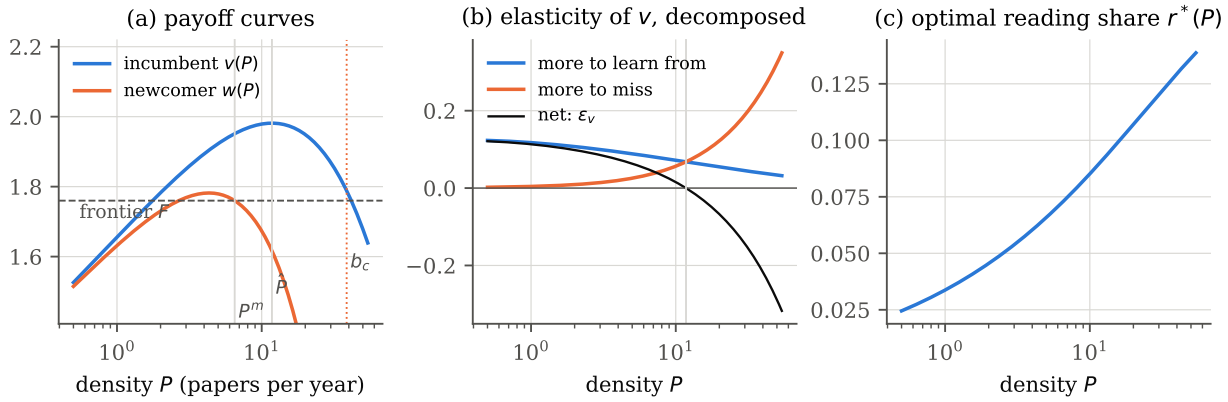


Figure 1: The payoff curve and its two forces. Left: $v(P)$ and the newcomer's payoff $w(P) = v(P)(1 - P/b_c)^\nu$ in the calibrated model, with the peak $\hat{P}$, the mature density P^m where $w = F$, and the speed limit b_c at which w vanishes. Right: the decomposition (12) of v'/v into the learning and overlap terms, and the optimal reading share.

of her time ($s_r = r^*$ for reading, $s_x = x^*$ for production) by $\mathrm{d}\ln(\textit{speed})$ *saves* $s_k\,\mathrm{d}\ln(\textit{speed})$ *of her time and raises her payoff by* $s_k\,\mathrm{d}\ln(\textit{speed})/x^*(P)$. *Per unit of time saved, every speed-up is worth the same,* $1/x^*(P)$.

The proof is two lines (Appendix A). By the envelope theorem $\partial \ln v/\partial \ln b = r\,\partial_R \ln(QC)\,b$, and the first-order condition sets $b\,\partial_R \ln(QC) = 1/(1-r)$. A 30% faster reader in a topic where reading takes 7% of time therefore gains about two log points, and a 30% faster producer about 26. The difference is the share of time each task takes, not anything about entry. It is entry that ranks tools whose value per hour saved is the same, and Section 6 shows how.

The sign of the spillover. Let one more paper appear in a topic. By the envelope theorem, the change in every other member's payoff is exactly $a\,v'(P)$. The researchers' own reallocation of time is second order.

Proposition 1 (Spillover sign)**.** *An additional paper in a topic of density P changes the per-period payoff of each incumbent by $a\,v'(P)$. The change is positive for $P < \hat{P}$ (more to learn from) and negative for $P > \hat{P}$ (more to miss).*

The sign of the spillover is therefore *derived*, not assumed. A paper helps its topic when the topic is thin and hurts it when the topic is dense. Overproduction is not built in. Whether science is overcrowded depends on where equilibrium places topics on v. That is the subject of the next two sections.

4 Free topic choice without frictions

We first set $b_c = \infty$, so $w = v$. Newcomers move freely between topics and the frontier. This benchmark isolates the topic-choice margin.

Proposition 2 (Equilibrium and dissipation)**.** *Let Assumptions 1 to 4 hold and $b_c = \infty$, and suppose $\hat{v} > F(N/n(\hat{P}), \hat{P})$. Then a topic equilibrium exists and is unique. Every open topic sits on the falling side of the payoff curve, $P^* > \hat{P}$, where an additional paper lowers the payoff of every member (Proposition 1). Output per researcher equals the frontier value, $a\,F < a\,\hat{v}$. The gain from cumulative knowledge, $\hat{v} - F$, is competed away. With founders who stay and $\eta_F = 0$, the equilibrium density solves $v(P^*) = e$, whatever the founders' probability ω.*

Entry continues past the peak because a newcomer compares the *average* payoff in a topic with the frontier. She ignores that her papers lower everyone else's payoff once the topic is past $\hat{P}$. Crossings on the rising side are unstable: one more entrant raises the payoff of entering. So topics fill past the peak, to the crossing on the falling side. Founders do not change this. They earn the same $v(P^*)$ as incumbents, so indifference between founding and joining pins $v(P^*)$ to what explorers produce.

Remark 3 (Cities, clubs and topics). Propositions 2 and 4 have the structure of the theory of city sizes (Henderson, 1974) and of clubs (Buchanan, 1965; Scotchmer, 2002): a single-peaked average payoff in group size, free entry of new groups, stability only on the falling side, a planner at the peak of the average payoff, and an entry tax that decentralises it. Self-organised cities are too large for the same reason (Henderson and Becker, 2000). Topics are the cities, newcomers the migrants, and the frontier the hinterland where new cities are founded. Three things distinguish topics. The hump is not assumed but derived from the two jobs of reading (Theorem 1). The congestion is not in land or roads but in attention: the crowders are the readers, so whether an extra member helps or hurts is decided by where the topic sits on its payoff curve (Proposition 1). And the object the data measure, the duplication rate, is a function of density alone, which makes the invariance of Proposition 3 a testable restriction rather than a curiosity.

Volume opens new topics. The next result says how a larger community distributes itself.

Proposition 3 (Volume opens new topics)**.** *In the large-topic limit, equilibrium density depends on (N, a) only through the product Na, and*

$$\mathcal{E} \equiv \frac{\mathrm{d}\ln P^*}{\mathrm{d}\ln N} = \frac{\eta_F}{\eta_F\, D + |\varepsilon_w| + \theta_F\, \varepsilon_U}, \qquad D \equiv 1 - \varepsilon_x + \frac{\rho}{\delta}\,\frac{P^*/b_c}{1 - P^*/b_c} \geq 1, \tag{13}$$

where $\varepsilon_w = \mathrm{d}\ln w/\mathrm{d}\ln P < 0$, $\varepsilon_x = \mathrm{d}\ln x^*/\mathrm{d}\ln P \leq 0$ and $\varepsilon_U = \mathrm{d}\ln \bar{U}_F/\mathrm{d}\ln P$ are evaluated at P^* ($\varepsilon_U = \varepsilon_v$ in the stationary economy). New topics absorb a share $\mathrm{d}\ln M^*/\mathrm{d}\ln N = 1 - D\,\mathcal{E}$ of growth, and papers per researcher move with elasticity $\varepsilon_x \mathcal{E} \leq 0$. If the frontier is not depleted ($\eta_F = 0$), density, the overlap rate and the aware share do not depend on N, a or the cost of entering science.

Equation (13) holds for any b_c, and we use it again in Sections 5 and 6. Three remarks.

First, the overlap rate is *pinned by entry, not by volume.* When the community grows, newcomers open new topics until existing ones are no more attractive than the frontier. The flagged overlap

rate and the aware share are functions of P^* alone (Lemma 1), so they inherit its invariance. A flat overlap rate while submissions double (Fact 1) is what the model predicts. It is not evidence that crowding is harmless.

Second, a fixed field corresponds to $\eta_F \to \infty$. Then $\mathcal{E} \to 1/D$: growth thickens existing topics. The arXiv elasticities of 0.12–0.19 at broad radii and 0.035 (SE 0.118) at the closest-overlap radius, against fixed-field benchmarks of 0.53–0.81 (Fact 1), imply that most growth opens new ground. Inverting (13),

$$\eta_F = \frac{\mathcal{E}\left(|\varepsilon_w| + \theta_F \varepsilon_U\right)}{1 - \mathcal{E}\, D}, \tag{14}$$

so the frontier's depletion is identified by the volume elasticity, the slope of the newcomer's payoff curve and the founders' share. The inversion assumes that the frontier did not move for other reasons over the sample. A frontier that improved, for instance because founding got easier, would lower measured density growth and bias η_F down (Theorem 3).

Third, papers per author barely move when the community grows, because $\varepsilon_x \mathcal{E}$ is the product of two small numbers. Faster production, by contrast, raises papers per author one for one (Proposition 7). The two can be told apart only at the author level, because entrants change the composition of authors (Fact 5).

Remark 4 (Short-run and long-run elasticities). Proposition 3 compares steady states, while the arXiv elasticities are measured over 2018–2025, a period in which volume grew about 12% a year. In the model, mature topics adjust to a change in N at once, since the marginal newcomer is indifferent. The only wedge between the measured and the steady-state elasticity is composition. Faster growth means more young topics, which are thinner. If an area's growth rate rises by $\mathrm{d}g$ for T years, its age weights for topics younger than T scale by $e^{\mathrm{d}g(T-a)}$ and its log volume rises by $T\mathrm{d}g$, so

$$\mathcal{E}_{\text{measured}} = \mathcal{E} + \frac{\mathrm{Cov}_\omega\left(\ln P, (T-a)^+\right)}{T} + o(\mathrm{d}g), \tag{15}$$

where Cov_ω is the researcher-weighted covariance over the cross-section of topic ages a. The covariance is negative because young topics are thin, so the measured elasticity is a lower bound. In the calibrated life cycle the bias over a seven-year window is -0.064 (Section 9). The long-run elasticity is therefore at most about 0.064 above the measured one, which leaves the conclusion that volume mostly opens new ground intact. Steady growth at a constant rate does not bias the estimate at all, because area fixed effects absorb a constant age composition.

Remark 5 (Reading capacity per researcher or per field). Consider instead a fixed screening capacity B per field, so that coverage is B/P and researchers produce full time. Topic choice still pins density, so Propositions 2 and 3 survive. But if topics cannot split, overlap rises with field volume, which the arXiv volume test rejects. Per-researcher capacity also keeps the time-allocation margin that prices tools by the time they save (Lemma 5). We adopt it throughout.

The planner. A planner who controls how many researchers join each topic, but not how they split their time, maximises output subject to births equalling deaths and researchers adding up.

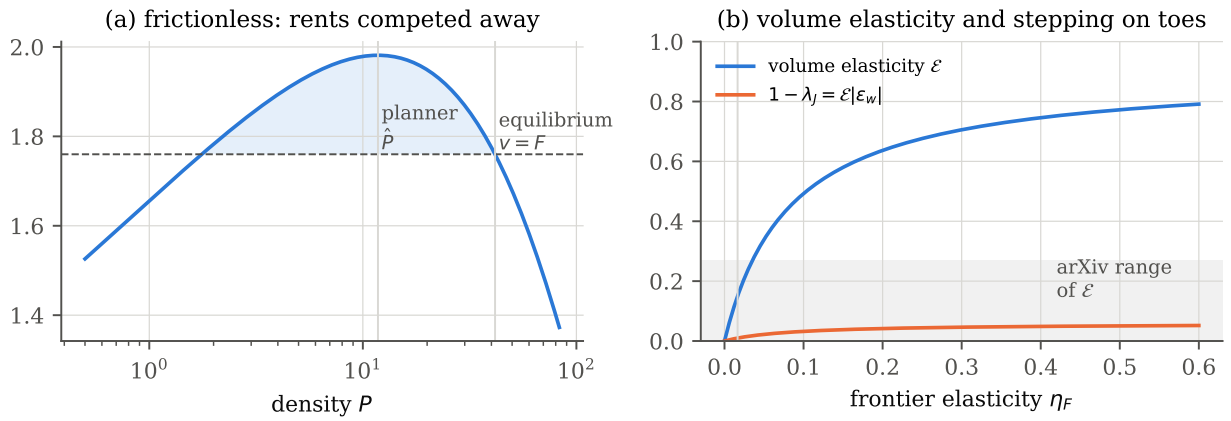


Figure 2: Left: without catch-up costs, topics fill to the density where $v = F$, and the shaded rent between the payoff curve and the frontier is competed away; the planner puts topics at the peak. Right: the volume elasticity $\mathcal{E}$ and the stepping-on-toes wedge $\mathcal{E}|\varepsilon_w|$ as functions of the frontier's depletion η_F, at the calibrated slope of w and founders' share. The shaded band is the range of arXiv estimates of $\mathcal{E}$.

Proposition 4 (Overcrowding and the frontier subsidy)**.** *Let the conditions of Proposition 2 hold, with $b_c \leq \infty$ and $\eta_F = 0$.*

(i) In the large-topic limit output per researcher at any common density is a $w(P)$, so the planner puts every topic at the peak of the newcomer's payoff curve, $P^o = \hat{P}_w$, which is the peak of v when $b_c = \infty$. She opens more topics and puts fewer researchers in each. The gain in output per researcher is $\hat{w}/F - 1$.

(ii) With explorers counted and founders who do not stay ($\omega = 0$), the planner's density satisfies

$$\frac{w(P^o) - e}{w(P^o)} = |\varepsilon_w(P^o)| \, \frac{1 + s\, n(P^o)/\xi}{D(P^o)}, \tag{16}$$

with $\hat{P}_w < P^o < P^*$. Overlap stays positive at the optimum.

(iii) In the stationary economy a frontier subsidy (equivalently, a tax on joining open topics) of $s^o = \hat{w} - F(M^o, \hat{P}_w)$ per unit of production speed decentralises the large-topic optimum.

(iv) (Two wedges) A planner who also chooses x wants researchers to produce less than $x^*(P)$ wherever $w'(P) < 0$ and more wherever $w'(P) > 0$. The production wedge is proportional to $(\ln w)'(P)$. In the large-topic limit it vanishes at the optimal density, so controlling topic choice alone attains the first best.

There are two wedges, and they differ in importance. Newcomers choose topics on the average payoff while the planner uses the marginal one: this is the *crowding wedge*. Inside each topic, researchers ignore that their papers add to what colleagues must read: this is the *production wedge*. Part (iv) shows the production wedge is a symptom of the crowding wedge. Once topics are at their best density, researchers' private time allocation is efficient. Part (iii) holds in the stationary

economy. In an economy where topics are born thin and grow, a flat subsidy to the frontier also deters the first followers of newborn topics, and the policy has to depend on a topic's age (Proposition 11). The welfare criterion here is total private credit; Proposition 6 shows that the overcrowding result survives when part of that credit is a transfer, and Proposition 8 values output by readership instead.

5 Catch-up, the speed limit and rents

The frictionless benchmark predicts equal returns across topics. The arXiv data reject this (Fact 2). The same researcher earns more citations in denser topics over most of the density range, and less in the densest ones. We now let $b_c < \infty$: newcomers must read a topic's stock before contributing, while its literature keeps growing.

5.1 Mature topics

Proposition 5 (Mature topics under catch-up costs)**.** *Let Assumptions 1 to 4 hold and $b_c < \infty$.*

(i) *$w(P) = v(P)(1-P/b_c)^\nu$ is strictly log-concave on $(0, b_c)$, with a unique peak $\hat{P}_w < \min\{\hat{P}, b_c\}$, and vanishes at b_c.*

(ii) *In a topic equilibrium the mature density P^m solves $w(P^m) = F$ on the falling side of w, and $P^m < b_c$. Incumbents earn $a\,v(P^m)$ per period. Their log premium over the frontier equals the catch-up cost, $\ln[v(P^m)/F] = -\nu \ln(1 - P^m/b_c) = (\rho + \xi)\, T(P^m)$.*

(iii) *Mature topics lie on the* rising *side of v if and only if catch-up is costly relative to the rent at the peak: $\hat{P} \geq b_c$, or $\nu \ln(1 - \hat{P}/b_c) < \ln(F/\hat{v})$.*

(iv) *Let topics differ in fertility ζ, so that payoffs are $\zeta v(P)$, and let $P^m(\zeta)$ solve $\zeta w(P) = F$. Across mature topics, log returns rise with log density with slope equal to the catch-up elasticity,*

$$\frac{\mathrm{d} \ln[\zeta\, v(P^m(\zeta))]}{\mathrm{d} \ln P^m(\zeta)} = \varepsilon_\Gamma(P^m) \equiv \nu\, \frac{P^m/b_c}{1 - P^m/b_c},$$

which exceeds the premium $-\nu \ln(1 - P^m/b_c)$.

Catch-up turns the dissipated rent into a protected one. Incumbents earn more than the frontier, and the premium exactly offsets what a newcomer gives up while she reads her way in. More fertile topics attract more entrants, have larger stocks and so protect larger premia. This yields a positive returns–density gradient across topics, whether those topics lie left or right of $\hat{P}$. In the data the same gradient also reflects audience size, since a paper with more close neighbours has more potential citers. The measured gradient of 6.7% per standard deviation (Fact 2) is therefore an upper bound on the catch-up channel. We use it as such in Section 9.

5.2 The speed limit

Theorem 2 (Speed limit)**.** *Let $b_c < \infty$.*

(i) (Closure.) *A topic whose flow is at least b_c is closed to newcomers: anyone who joins it while its flow stays there never catches up. Its incumbents' rents are protected however large they are.*

(ii) (Rents near the limit.) *Every mature topic satisfies $P^m < b_c$. As fertility rises, $P^m(\zeta) \to b_c$, and both the incumbents' premium and the cross-topic returns gradient diverge; the gradient grows faster than the premium.*

(iii) (Growing topics.) *Myopic entry into a topic that grows stops before its flow reaches b_c, however fast it grows.*

(iv) (Production shocks close topics.) *A production speed-up by the factor $1+\alpha$ raises the flow of every topic on impact, before membership adjusts, by the same factor at given reading shares. Every topic with $P^m > b_c/(1+\alpha)$ closes, and stays closed for at least $\ln[(1+\alpha)P^m/b_c]/\rho$ periods while turnover thins it.*

The theorem makes "too fast" precise. A literature can grow faster than the people outside it can absorb, and when it does, the topic is shut to newcomers however rewarding it is for those inside. Near the limit, small differences in fertility produce large differences in returns, so the speed limit is also where the returns gradient is steepest. Part (iv) is the version that matters for artificial intelligence. A tool that lets everyone produce faster raises every topic's flow at once, and the topics that were already densest are pushed past the limit until turnover brings them back. Entry, and the turnover of people and ideas that comes with it, stops there for a while. This is one mechanism behind the slowed progress of large fields documented by Chu and Evans (2021): when papers arrive faster than newcomers can absorb them, attention stays with the established canon. Better catch-up tools, such as reviews and summaries, raise the limit.

5.3 Rising incumbent returns, falling newcomer returns

The interesting object is the gap between what incumbents earn, v, and what a newcomer can expect over her spell, $w = v\Gamma$. Incumbents never pay the catch-up cost again. Newcomers and the planner do. The welfare statement follows almost by definition of stability, so we state it as a corollary and put the weight on what it implies for measurement.

Corollary 1 (Overcrowding with rising returns)**.** *In every stable topic equilibrium, one more member in a topic of density P changes total output of the other members by $n\,a\,w'(P)\,\mathrm{d}P/\mathrm{d}n < 0$. This counts both the spillover on incumbents (v') and the extra catch-up she imposes on future newcomers ($v\,\Gamma'$). If $w(\hat{P}) < F$, then $\hat{P}_w < P^m < \hat{P}$: incumbents' returns rise with density ($v'(P^m) > 0$) while the topic is too crowded ($w'(P^m) < 0$). The loss of output per researcher in the large-topic limit is $1 - F/\hat{w}$.*

The measured gradient is the incumbents' private return to density. A newcomer values a topic net of the reading she must do to join it, and so does the planner. Rising incumbent returns are the footprint of the catch-up rent, not of an untapped spillover. A natural reading of Fact 2 is that most researcher-years sit where more density raises returns, so most topics are too thin. The corollary says that this reading conflates v with w. Two kinds of topic can be too thin. The first is young topics that have not yet filled. The second, if the frontier is attractive enough, is newborn topics that never find followers (Remark 6).

Remark 6 (Newborn topics). A newborn topic has no stock, so its first follower needs no catch-up and earns $v(P_0)$, where P_0 is the flow of a newborn topic. Newborn topics attract followers if and only if $v(P_0) > F$. (The condition is on v, not on w: a stationary topic of density P_0 would carry a stock, a newborn one does not.) If $F > v(P_0)$, newborn topics attract no followers, although a coordinated group would earn $\hat{v} > F$. Science then holds too many topics that are too small, and the planner would fix it by seeding followers. When $v(P_0) > F > w(P_0)$, as in the calibration, a newborn topic is worth joining while its stock is empty, but a stationary topic of the same density is not. The topic must grow fast enough for its stock to lag behind its flow, or entry stalls before the topic is worth joining in its own right (Proposition 9). Policies that raise the frontier value by more than $v(P_0) - F$ shut down the birth of new topics (Proposition 11).

5.4 Social value and priority transfers

The welfare statements so far count private credit. Priority credit is partly a transfer (Merton, 1957; Hill and Stein, 2025a). When a later paper turns out to duplicate an earlier one, some of the credit it loses goes to the earlier paper, so part of the "loss" from overlap is redistribution. Conversely, a duplicate may be worth less to science than its residual citations suggest. We therefore separate social value from credit. Let the social value of a project be $Q\,C_s$, with

$$C_s(R, P) = \exp\{-\theta P[\lambda_s - \Delta_s \varphi(R/P)] - \theta_c \lambda_c P\},$$

where λ_s is the share of a project's social value lost per unread covering paper and Δ_s the part of it that detection averts. Researchers still choose $r^*(P)$ to maximise credit, and social output per unit of production speed in a topic of density P is $s(P) = (1 - r^*)\,Q(br^*, P)\,C_s(br^*, P)$, with $s_w(P) = s(P)\Gamma(P)$. The catch-up cost is a real resource cost, reading time, and is not a transfer.

Proposition 6 (Social value). *(i) One more member changes the social output of the others by* $n\,a\,s_w'(P)\,\mathrm{d}P/\mathrm{d}n$, *where*

$$\frac{s'(P)}{s(P)} = \frac{v'(P)}{v(P)} - \theta\big[(\lambda_s - \lambda) - (\Delta_s - \Delta)\psi(c^*)\big] + \theta b(\Delta_s - \Delta)\varphi'(c^*)\,r^{*\prime}(P),$$

with $\psi(c) = \varphi(c) - c\varphi'(c) \geq 0$.

(ii) (Uniform wedge.) *If* $\Delta_s = \Delta$, *then* $s(P) = v(P)e^{-\theta(\lambda_s - \lambda)P}$ *exactly. Social and private values then differ by a factor that rises or falls exponentially in density. The equilibrium is*

overcrowded relative to the social optimum if and only if the share of overlap losses that are transfers, $\tau \equiv (\lambda - \lambda_s)/\lambda$, satisfies

$$\tau < \frac{|\varepsilon_w(P^m)|}{\theta \lambda P^m}.$$

In particular overcrowding is stronger when duplicates are worth less to science than to their authors ($\lambda_s > \lambda$).

(iii) (Proportional transfer.) *If a share τ of every overlap loss is a transfer, $\ln C_s = (1-\tau)\ln C$, the social optimum density maximises $\ln v(P) - \tau \ln C^*(P) + \ln \Gamma(P)$, where $C^*(P) = C(br^*(P), P)$.*

Part (ii) gives a sharp test. Overcrowding survives any degree of transfer as long as the slope of the newcomer's payoff curve at the mature density exceeds the expected erosion intensity $\theta \lambda P^m$. In the calibrated model the slope is 0.065 against an erosion intensity of 0.043, so the critical transfer share is 1.5, above one. Under proportional transfers (part (iii)), the social optimum moves from 4.3 to 5.2 papers a year if all overlap losses are transfers, still below the mature density of 6.5. Transfers therefore shrink the welfare gain from optimal topic choice, from 1.2% to 0.4% of output per researcher, but they do not reverse its sign. The catch-up burden that newcomers impose on one another is a real resource cost, and on its own it keeps mature topics too dense. The transfer critique matters for magnitudes, and especially for the frictionless benchmark, where the dissipated rent is a difference in private payoffs; it does not matter for the direction of the inefficiency.

6 The incidence of research tools

New tools, artificial intelligence among them, enter the model through production speed a, reading speed b, learning β, search precision σ, the cost of pivoting κ_p, catch-up speed b_c, and the cost of entering science. Inside a topic, Lemma 5 prices each speed-up by the time it saves. Where that value ends up once researchers move is a question of incidence. Lemma 4 answers it: in the long run, output per researcher is the value of the frontier.

Theorem 3 (Frontier incidence)**.** *Let a tool raise $\ln w$ by h and $\ln F$ by f at a given density, where f includes the frontier's response to any change in the number of topics needed to hold the researchers at that density. In the large-topic limit,*

$$\mathrm{d}\ln \frac{Y}{N} = \mathrm{d}\ln a + \lambda_F f + (1-\lambda_F)h, \qquad \mathrm{d}\ln P^* = \frac{h-f}{\Psi}, \qquad \lambda_F \equiv \frac{|\varepsilon_w|}{\Psi}, \quad \Psi \equiv |\varepsilon_w| + \theta_F \varepsilon_U + \eta_F D. \tag{17}$$

In the stationary economy ($\varepsilon_U = \varepsilon_v$) the share of a newcomer-side gain that survives entry is

$$1 - \lambda_F = 1 - (1 - \mathcal{E}D)\,\frac{|\varepsilon_w|}{|\varepsilon_w| + \theta_F \varepsilon_v}, \tag{18}$$

which is at least $\mathcal{E}D$ when mature topics lie on the rising side of v.

In the long run a tool is worth what it adds to the value of opening a new topic. A tool that raises payoffs only inside existing topics ($h > 0$, $f = 0$) draws newcomers in until those topics are again no better than the frontier. Its gain survives only through the two channels by which the frontier depends on density: founders who stay earn the payoffs of the denser topics they end up in ($\theta_F \varepsilon_U$), and fewer topics leave better directions open at the frontier ($\eta_F D$). Equation (18) puts a floor under the surviving share that the data can see: the volume elasticity of density, scaled by $D \approx 1$. Without founders the floor is exact.

Corollary 2 (Ranking tools)**.** *In the large-topic limit:*

(i) Production tools *(a) act like a larger community: $f = -\eta_F \, \mathrm{d}\ln a$ and $h = 0$, so*

$$\mathrm{d}\ln(Y/N)/\mathrm{d}\ln a = 1 - \mathcal{E}|\varepsilon_w|.$$

(ii) Learning tools *(b, β, σ, κ_p) raise $\ln v$ by h_v at every density. Founders read too, so $f = \theta_F h_v$ up to the small change in the number of topics, and the share that survives is $\Pi_L = 1 - \lambda_F(1 - \theta_F)$.*

(iii) Catch-up tools *(b_c) raise only newcomers' payoffs. Founders never catch up, so $f = 0$ up to the same term, and $\Pi_C = 1 - \lambda_F$.*

(iv) Hence $\Pi_C \le \Pi_L \le 1 - \mathcal{E}|\varepsilon_w|$ whenever $\eta_F \le 1 - \theta_F$. At fixed topics every speed-up is worth the same per unit of time saved (Lemma 5); in the long run, per unit of time saved, a production tool is worth $(1 - \mathcal{E}|\varepsilon_w|)/\Pi_L$ times a reading tool.

The theorem replaces the dissipation result of earlier versions of this paper, which had $\theta_F = 0$ by construction: pioneers who never read and never stayed in the topics they opened, so that nothing that happened inside topics could reach the frontier. Once founders stay, reading tools reach the frontier directly, because founders read. Catch-up tools reach it only indirectly, through the density of the topics founders end up in, because founders never have to catch up. The asymmetry that survives is between tools that help the people who open topics and tools that help only those who join them. How large it is depends on θ_F, the share of the frontier's value that founders earn as incumbents, which is a measurable object: the share of an explorer's expected output that comes from her years as the founder of a topic.

Two further channels can be read off the theorem. If explorers need search to check that a direction is not already taken, so that their founding rate rises with search precision at elasticity $\varepsilon_{s\sigma}$, search tools also raise f by $(\theta_F - \omega)\, \varepsilon_{s\sigma} \, \mathrm{d}\ln\sigma$, since $\partial \ln F / \partial \ln s = \theta_F - \omega$. And on the life cycle two qualifications apply that the stationary theorem does not capture. Founders spend much of their tenure in topics that are still growing at a pace set by diffusion, so their payoff responds less to the mature density than ε_v suggests. And entry into a growing topic is limited by diffusion, so its members earn rents that entry does not compete away. Section 9 quantifies both on the simulated life cycle.

Proposition 7 (Footprints of new research tools)**.** *In the large-topic limit:*

(i) Faster production *($a \uparrow$) acts like a larger community. Density moves with elasticity $\mathcal{E}$ (zero if $\eta_F = 0$) and the number of topics with elasticity $1 - D\mathcal{E}$. Papers per researcher rise with elasticity $1 + \varepsilon_x \mathcal{E}$, and output per researcher with elasticity $1 - \mathcal{E}|\varepsilon_w|$.*

(ii) Cheaper entry into science *($c \downarrow$, with N set by $aF = u_0 + c$ for an outside option u_0, and $\eta_F > 0$) raises N. Density rises with elasticity $\mathcal{E}$ with respect to N, papers per researcher move with elasticity $\varepsilon_x \mathcal{E} \approx 0$, and new topics absorb a share $1 - D\mathcal{E}$ of growth.*

(iii) Better reading, learning or search, or cheaper pivots *(b, β or $\sigma \uparrow$, $\kappa_p \downarrow$) raise v and w at every density, and F by the founders' share of the gain. Since $\theta_F < 1$, density rises and the number of topics falls: science concentrates.*

(iv) Faster catch-up *($b_c \uparrow$, for example review articles or literature-summarising tools) raises w at every density and leaves founders' payoffs unchanged, with the effects in (iii). It also raises the speed limit and lowers the overshooting threshold of Proposition 9, so more topics grow too fast.*

(v) Better search *($\sigma \uparrow$) raises the aware share A at any given reading coverage.*

Parts (iii) and (iv) follow from the envelope theorem. Each parameter raises $g(r, P)$ at every (r, P), hence raises v, or raises Γ. A pointwise upward shift of a single-peaked curve moves its falling-side crossing with the frontier to the right whenever the frontier rises by less. The effect on the measured overlap rate is ambiguous in general: denser topics generate more overlap, but better detection removes more of it.

Readership. The welfare criterion so far is credited output: papers weighted by their quality and credit. Reads per paper are $\mathcal{A} = (b/a)\, r^*/(1 - r^*)$, since members read $b\, r^*$ papers each while producing $a\, x^*$. A faster pen with unchanged reading means each paper is read less.

Proposition 8 (Valuing output by readership)**.** *Let output be valued as $Y_\alpha = \mathcal{A}^\alpha Y$, where $\alpha \in [0, 1]$ is the weight of readership. Then $\mathrm{d}\ln(Y_\alpha/N) = \mathrm{d}\ln(Y/N) + \alpha\, \mathrm{d}\ln \mathcal{A}$, and at a given density $\mathrm{d}\ln \mathcal{A} = \mathrm{d}\ln b - \mathrm{d}\ln a + \mathrm{d}\ln[r^*/(1 - r^*)]$. Valued by readership ($\alpha = 1$), a production tool raises output per researcher by $-\mathcal{E}|\varepsilon_w|$ plus the change in $r^*/(1-r^*)$ that comes with density, approximately nothing, while a reading tool raises it by approximately $\mathrm{d}\ln b$. The ranking of Corollary 2 reverses.*

Total reading is fixed by researchers' time, and a production tool spreads it over more papers. If science values papers by how much they are read, only reading tools raise output, and they raise it nearly one for one. If it values them by what they contribute whether or not they are read widely, the ranking of Corollary 2 applies. Citations sit between the two: they are given by writers, so total citations grow with the number of papers, but which papers are cited depends on what was read. We report both units. Appendix B.5 adds a third qualification. When search for overlap must be repeated for every project, a production speed-up is worth only the production share of project time, and it lowers the search done per project, so duplicates rise.

Table 2: The incidence of new research tools in the calibrated model

Tool	Shock	Output per researcher			Footprints ($\omega = 0.5$)				
		Fixed topics	Long run $\omega = 0$	Long run $\omega = 0.5$	Density	Topics	Papers/ researcher	Flagged overlap	Aware (pp)
Faster production	$a \times 1.3$	+26.2	+26.0	+26.0	+3.8	+22.0	+26.1	+5.0	−1.2
Faster reading	$b \times 1.3$	+2.0	+0.3	+1.5	+7.8	−8.0	+0.5	+1.8	+6.4
Better learning	$\beta \times 1.3$	+4.1	+0.5	+2.9	+14.6	−16.3	−0.4	+19.1	−4.6
Better search	$\sigma \times 1.3$	+0.1	+0.0	+0.1	+0.5	−0.4	+0.1	−11.2	+12.9
Cheaper pivots	$\kappa_p \div 2$	+0.6	+0.1	+0.5	+2.5	−3.0	−0.2	−35.9	−18.3
Faster catch-up	$b_c \times 1.3$	+2.6	+0.4	+1.0	+26.3	−27.1	−0.8	+34.6	−8.3

Stationary economy, solved exactly. Log points (×100) of a 30% shock (pivot cost halved). *Fixed topics*: change in the newcomer's payoff w at the mature density, which for a speed-up equals the time it saves (Lemma 5). *Long run*: change in output per researcher aF, with the frontier's depletion η_F set so that the volume elasticity is 0.15, for explorers who never stay ($\omega = 0$) and for founders who stay in the topics they open with probability ω. The flagged overlap rate is relative to its baseline level, which is a normalisation.

Table 2 reports the calibrated magnitudes. The proposition matches Evans (2008), who finds that online access narrowed the set of papers that get cited, and Hao et al. (2026), who find that AI tools expand individual scientists' impact but contract science's collective focus. Gains from access to frontier knowledge measured for individuals or topics (Iaria et al., 2018; Biasi and Moser, 2021) are real, but in equilibrium they reach science as a whole only through the frontier.

What the data can and cannot yet say. Sections 7 and 8 take the footprints to the arXiv record. At the author level, adopters of LLM-style writing raised their output by about 7% relative to comparable non-adopters, also in fields where AI is not the subject of research (Fact 5). Because a is research output per unit of production time, a tool that speeds up drafting raises a only by the share of production time that goes into drafting, so a small effect on output is what the model expects from a drafting tool. The area-level evidence on pioneers and on overlap measured by meaning cannot separate channels: the first is confounded by changes in wording and the second is underpowered (Section 7). The reading and catch-up channels remain predictions. The model says that when literature-summarising tools become widespread, topics will get denser, fewer new ones will open, more fast-growing topics will overshoot, and output per researcher will rise by much less than the time the tools save, unless founders are the ones who use them. The arXiv rule that took most review articles out of computer science in November 2025 is a first shock to catch-up costs. Section 8 reports its first stage, and Section 11 states what the model predicts it will do.

7 Evidence I: the arXiv record

This section documents five facts that discipline the model. The analysis frame holds 1.49 million arXiv papers posted in 2018–2025 (1.65 million through mid-2026), classified into 150 text-based fine areas. Volume rose by 122% over 2018–2025. It is complemented by a sample of 25,400 full-text PDFs for citation behaviour, by citation counts and reference lists, and, for the validation of

awareness, by curated INSPIRE-HEP reference lists. A paper's *overlap flag* is set when an earlier paper by another group, posted 3 to 24 months before, lies above its area's 99th percentile of 2018–19 nearest-neighbour TF-IDF similarity. The *level* of the flagged rate, about 1% in the base period, is a normalisation; its *changes* carry the information. Text-based similarity follows Hoberg and Phillips (2016) and Kelly et al. (2021). TF-IDF and sentence-embedding flags agree on only about a fifth of flagged papers, so we report which measure each fact uses and do not lean on either alone. Table 3 collects the numbers. Standard errors are clustered by author unless noted. Clustering by area would be more conservative and is part of the replication package.

Fact 1. *Volume opens new ground.* We regress the log local-neighbour hazard, the rate at which a paper's similarity radius is populated by earlier papers, on the log of the area's prior two-year volume with area and year effects. From the 95th to the 25th percentile radius, the elasticity is 0.12–0.19. If an area's papers kept landing in the same idea space, randomly thinning each paper's observed neighbour list implies 0.53–0.81 at the same radii. At the 99th percentile radius, the radius of the overlap flag, the estimate is 0.035 with a standard error of 0.118 (95% interval -0.20 to 0.27). In the all-arXiv time series, volume rose 122% from 2018 to 2025 while the overlap hazard rose about 13% (elasticity 0.19–0.27). The model's topic is a cell of fixed size in idea space (Remark 1), so its density is the close-radius density. We therefore use $\mathcal{E} = 0.15$ as the baseline and report results over $[0, 0.27]$. Broad radii fill faster than close ones, as new topics are packed into an area's idea space (prediction P7 in Section 11).

Fact 2. *Returns are hump-shaped in topic density.* Comparing the same researcher across years spent in denser or sparser topics, with author and year effects (923,000 author-years), one standard deviation of density goes with 6.7% more fractional citations per author-year (SE 0.15 percentage points, clustered by author). Relative to the middle fifth of density, returns are -11% in the sparsest fifth, $+2$% in the fourth fifth and -2% in the densest. The densest researcher-years thus earn less than the middle. A quadratic puts the peak about one standard deviation above mean density, with 12% of researcher-years beyond it. The shape survives using the third-closest paper and restricting to researchers active three or more years. Two caveats matter for interpretation. First, citations reward audience. A paper with more close neighbours has more potential citers, which by itself produces a rising gradient, so the rising arm is an upper bound on the returns to density. Second, papers per author-year rise slightly with density within author, $+0.5$% per standard deviation (SE 0.1). Theorem 1(v) predicts a fall. The discrepancy is small, but it is a failed prediction and we report it. Larger teams in denser topics are one candidate explanation.

Fact 3. *Duplicates lose some credit, aware follow-ons gain.* Among 2018–2021 papers with at least five references, relative to papers in the same area-year with the same topic density, *unaware* overlaps (which do not cite their nearest earlier paper) receive 19% fewer citations (SE 2). *Aware* overlaps (which cite it) receive 29% more (SE 5). The penalty is close to

Table 3: The arXiv facts

Fact	Statistic	Value
1	Elasticity of local density to area volume, 95th–25th pct. radius	0.12–0.19
	at the overlap radius (99th pct.)	0.035 (0.118)
	fixed-field benchmark at the same radii	0.53–0.81
	all-arXiv time series, 2018–2025 (volume +122%)	0.19–0.27
2	Citations per author-year per SD of density (author FE)	+6.7% (0.15 pp)
	By fifth, relative to the middle: sparsest / 4th / densest	−11% / +2% / −2%
	Peak of quadratic; share of researcher-years beyond it	≈ +1 SD; 12%
	Papers per author-year per SD of density	+0.5% (0.1)
3	Citations: unaware overlap / aware overlap	−19% (2) / +29% (5)
4	Aware share of close overlap: raw / recall-corrected	35–42% / ≈ 60%
	INSPIRE-HEP, 90th / 75th pct. similarity	61% / 50%
5	Author-level output of LLM-style adopters, 2024–26	≈7%

Notes: arXiv 2018–2025, 150 areas. Standard errors in parentheses. Facts 1–4 discipline the calibration of Section 9. Fact 5 bears on the drafting channel of Proposition 7.

the 21% citation penalty for scooped teams in structural biology estimated by Hill and Stein (2025a) with a different design and field. The two estimates validate each other.

Fact 4. *About 60% of close overlap is aware.* In the PDF sample, the raw share of flagged pairs in which the later paper cites the earlier one is 35–42% in every year from 2019 to 2026. PDF matching recovers 63% of citations, so the recall-corrected share is about 60%. Recall is not constant: arXiv identifiers recovered per PDF rose from 5.0 in 2019 to 8.8 in 2026, and the raw aware share rose by 3.8 percentage points (SE 1.3) after 2022. Only the comparison across areas with high and low exposure to large language models is flat (−0.5 pp per standard deviation, SE 1.4). An independent check comes from high-energy physics, where INSPIRE-HEP curates complete reference lists. Among later papers by other authors that overlap an earlier paper at the area's 90th percentile of nearest-neighbour similarity, 61% cite it (50% at the 75th percentile). We calibrate to 60%.

Fact 5. *Growth came mostly from new authors, and drafting tools raised output per adopter only modestly.* At the area level, papers per unique author identifier stayed flat while the number of authors grew. An area-level ratio cannot rule out a drafting effect, because entrants change the composition of authors. Author identifiers are also unreliable in 2025, when many were reissued. We therefore build an author-level panel with name-keyed submitter identities (Section 8). Submitters whose 2023 abstracts adopt the vocabulary that spread with large language models (Liang et al., 2024; Kobak et al., 2025) raised their output in 2024–2026 by about 7% relative to submitters who did not, and by a similar amount in fields where AI is not the subject. Kusumegi et al. (2025) report 36% more papers for LLM adopters. Their timing rule, which dates adoption by the first flagged paper, mechanically generates positive dynamics (Renault et al., 2026). Classifying adoption before the outcome window, as we do, avoids this.

What we no longer claim. An earlier version of this evidence reported that areas with high LLM uptake produced more pioneer papers, and that overlap measured by meaning did not fall. Neither survives scrutiny. The pioneer measure uses the same TF-IDF neighbours that changes in wording distort. In high-uptake areas, rare words in abstracts rose from 25.4 to 33.0 per abstract and hyphenated words from 6.9 to 10.5 over 2025–2026. Without detrending against a pre-existing decline, the pioneer estimate is −0.8 pp (SE 0.2), which points to concentration rather than dispersion. The embedding-based overlap estimate, −0.19 pp (SE 0.24) on a base near 1%, has a minimum detectable effect at 80% power of about 0.67 pp, roughly two thirds of the base. Most model channels move flagged overlap by 7–30% (Table 2), so the test cannot discriminate. Finally, 13 of the 15 highest-uptake areas study AI or computer vision, so area exposure confounds AI as a tool with AI as a subject. The author-level design in Fact 5 addresses this by restricting to fields where AI is not the subject.

Mapping to the model. Fact 1 is the volume elasticity $\mathcal{E}$ of Proposition 3. Fact 2 rejects the equal returns of the frictionless model and motivates the catch-up cost of Section 5. Fact 3 disciplines the credit parameters λ and Δ/λ (Lemma 1), and Fact 4 disciplines search precision σ through the aware share (4). None of these facts is causal. The next section turns to designs that are.

8 Evidence II: arXiv's rules as experiments

The facts above are correlations. This section uses arXiv's own rules, which create variation in how much a paper is read, in the cost of entering, in the cost of catching up on a literature and in how much one person may write. The data are the arXiv metadata snapshot of 18 October 2026 (Cornell University, 2026): 2.2 million papers with new-style identifiers submitted from 2015 to 15 October 2026, with the exact time of each submission, the submitter's name, the author list and the abstract. Appendix D documents every construction. Tables 4 and 5 report the estimates.

Reconstructing the daily listings. arXiv announces new papers once a day. Papers submitted before the daily cutoff appear in that evening's listing, and later papers wait for the next one. Identifiers are assigned at announcement, in listing order. Within an announcement batch, the rank correlation between identifier and submission time is 0.98 for papers that were not held for moderation (median across batches). Six percent of papers are held and announced later. The submission-time distribution reveals the cutoff. It was 16:00 Eastern Time until the end of 2016 and has been 14:00 since January 2017. On an average weekday about 15 papers are submitted in the first ten seconds after 14:00:00: authors race for the top of the next listing (Haque and Ginsparg, 2009). We place every paper in its announcement batch and rank it within its primary category's list of new submissions.

8.1 Does reading protect? Listing position as a readership shock

Lemma 1 gives sharp predictions for a shock that makes a paper i more visible to the researchers who might later cover it. If visibility raises the probability φ that a later covering project detects i, unaware duplicates of i fall by $\theta\,\mathrm{d}\varphi$, aware follow-ons that cite i rise by $q\theta\,\mathrm{d}\varphi$, and the ratio of the two effects identifies q, the share of detections that proceed. We test these predictions with two designs.

Design A: the daily cutoff. A paper submitted just after 14:00 opens the next listing, while one submitted just before closes the current one. The first stage is sharp (Table 4, panel A; Fig. 3). Within an hour of the cutoff, excluding a five-minute donut around the bunching, crossing the cutoff raises the probability of being among the first five papers by 0.69, from a base of 0.30. The comparison it delivers, however, is first against *last*. The end of a listing is also highly visible, a "reverse visibility" effect documented for arXiv by Haque and Ginsparg (2010) and for NBER working papers by Feenberg et al. (2017). Submitters on the two sides also differ: those just after the cutoff have fewer prior submissions (-0.051, SE 0.017). Top-of-listing papers receive *fewer* citations than end-of-listing papers: -0.137 log points (SE 0.060) in high-energy physics, where INSPIRE-HEP gives complete citation counts, and -0.175 (SE 0.032) across all fields in the 43,097 papers with Semantic Scholar counts. The cutoff therefore does not isolate visibility, and we do not use it to test detection.

Design B: rank within the listing. A cleaner source of variation is how many other papers happen to be submitted earlier on the same day. A paper's position is one plus the number of papers in its category submitted since the batch opened. We hold fixed the submission time (in 15-minute bins, separately for the weekend batch), the category-year, the announcement date and the size of the listing. Position then varies only with the timing of other people's submissions. We use all 87,694 papers in the nine categories covered by INSPIRE-HEP in 2017–2022, excluding the first 15 minutes after the cutoff. The residual standard deviation of log position is 0.45. Predetermined characteristics are balanced: number of authors and the submitter's prior submissions are unrelated to residual position. Abstracts are slightly shorter lower in the list (1.6 words of 151 per log point, SE 0.6).

The readership effect is small. One log point lower in the listing, roughly moving from second to fifth place, reduces citations by 0.023 log points (SE 0.016; 0.031 without date fixed effects). Position bins show no premium for the very first slots once dates are held fixed. Later overlapping work responds even less. We count later papers in the same category, within 24 months, by authors who do not overlap with the focal paper's, whose TF-IDF similarity exceeds the category's median (75th percentile) of nearest-neighbour similarity. With INSPIRE's curated reference lists we split them into those that cite the focal paper (aware) and those that do not (unaware). One log point lower in the listing changes unaware overlap at the 75th percentile threshold by -0.0049 (SE 0.0038) on a mean of 0.068, and aware overlap by -0.0001 (SE 0.0041). The 95% interval rules out an

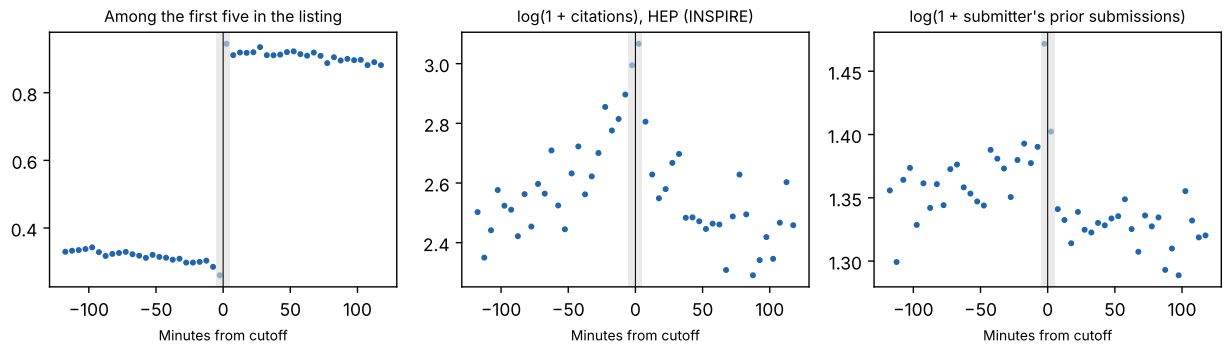


Figure 3: The daily cutoff. Binned means, net of category-year means, by minutes from the 14:00 ET cutoff (weekday submissions 2017–2022; shaded: five-minute donut). Left: probability of being among the first five papers of the category's listing. Middle: log citations in high-energy physics (INSPIRE-HEP). Right: the submitter's prior submissions, a predetermined characteristic that jumps at the cutoff. Papers just after the cutoff open the next listing; papers just before close the current one.

increase in unaware overlap of more than 3.9% per log point. The minimum detectable effect at 80% power is 16% of the mean. Self-follow-ups by the focal paper's own authors fall with position (−0.048, SE 0.015). This suggests that attention encourages authors to continue a line of work, or that the design leaves some day-level confounding; either way it does not affect the comparison with other authors' work.

What this teaches. Passive visibility moves casual readership a little and does not move the detection of overlapping work. In the model, detection depends on the coverage R/P that a later researcher chooses and on search precision σ, not on how prominently the earlier paper was announced. The result says that the protective job of reading is done by targeted search when a project is planned or written up, not by browsing daily listings. This matches the high calibrated search precision needed for a 60% aware share (Fact 4). The right causal test of φ is therefore a shock to the cost of search. Section 11 sets out one: the blocking of Google in mainland China from 2014 (Zheng and Wang, 2020). The model predicts that it lowered the aware share and raised unaware duplication in work by China-based authors, relative to Hong Kong-based and overseas Chinese authors in the same topics.

8.2 Entry cost: the endorsement rule

New arXiv submitters need an endorsement. Until late 2025, an e-mail address at a recognised institution sufficed. On 10 December 2025 arXiv stopped accepting it alone in Mathematics (arXiv, 2025b), and on 21 January 2026 in all sections (arXiv, 2026a). New authors now need prior authorship of an accepted paper in the section or a personal endorsement. We measure the share of papers whose submitter has never submitted before. We compare each week with the same week of the previous academic year, which removes seasonality, and Mathematics with other sections,

Table 4: Listing position, readership and later overlapping work

Outcome	Estimate	(SE)	Mean	N
A. Discontinuity at the daily cutoff: submitted just after (top of next listing) vs. just before (end of current listing)				
Position in listing (0 = top, 1 = bottom)	−0.838	(0.004)	0.884	77,051
Among the first five papers	+0.688	(0.006)	0.300	77,051
Number of authors (balance)	−0.178	(0.454)	5.650	77,051
Abstract length, words (balance)	−0.832	(0.917)	153.785	77,051
log(1 + submitter's prior submissions) (balance)	−0.051	(0.017)	1.363	77,050
log(1 + citations), HEP (INSPIRE)	−0.137	(0.060)	2.682	11,483
log(1 + citations), all fields (Semantic Scholar)	−0.175	(0.032)	2.522	43,097
B. Listing rank in high-energy physics: effect of log position, holding submission time fixed				
log(1 + citations)	−0.023	(0.016)	2.309	87,679
Citation percentile in category-year	−0.005	(0.003)	0.480	87,679
Later overlapping papers by others, τ_{50}	−0.021	(0.018)	0.667	87,694
of which cite the focal paper (aware)	−0.003	(0.008)	0.211	87,694
of which do not (unaware)	−0.018	(0.014)	0.454	87,694
Later overlapping papers by others, τ_{75}	−0.005	(0.006)	0.130	87,694
aware	−0.000	(0.004)	0.062	87,694
unaware	−0.005	(0.004)	0.068	87,694
Later overlapping papers by the same authors, τ_{50}	−0.048	(0.015)	0.687	87,694

Notes: Panel A: local linear regression discontinuity at the 14:00 ET cutoff, bandwidth 60 minutes, triangular kernel, donut of 5 minutes, separate slopes, category-year and weekday fixed effects; standard errors clustered by submission day; "Mean" is the mean just before the cutoff. Panel B: all papers in hep-th, hep-ph, hep-ex, hep-lat, gr-qc, nucl-th, nucl-ex, astro-ph.CO and astro-ph.HE announced in their regular batch, 2017–2022, submitted at least 15 minutes after the batch opened; regressors are log listing position and log list size; fixed effects for category-year, submission-time bin (15 minutes; hourly for the weekend batch) and announcement date; standard errors clustered by announcement date. Overlap thresholds τ_q are percentiles of the category's nearest-neighbour similarity distribution. Citations and reference lists from INSPIRE-HEP (excluding self-citations) unless noted.

which exploits the staggered timing.

The first stage is large and well timed (Table 5, panel A; Fig. 4). Net of the year before, the first-time submitter share fell by 2.4 percentage points in Mathematics (SE 0.3; about 17% of its base) and by 3.4 points in other sections (SE 0.3; about 15%). In the six weeks when only Mathematics was treated, Mathematics fell 1.7 points more than the other sections (SE 0.4). Submitters with no prior arXiv authorship, the group the rule targets, fell by about 17% outside Mathematics. In the model, this is a rise in the cost of entering science. Proposition 7(ii) predicts that it reduces the number of topics roughly in proportion to the entrants it removes, while leaving density, the overlap rate and the aware share nearly unchanged, because $\mathcal{E}$ is small. Those outcomes need a year of post-period data; Section 11 states them in advance.

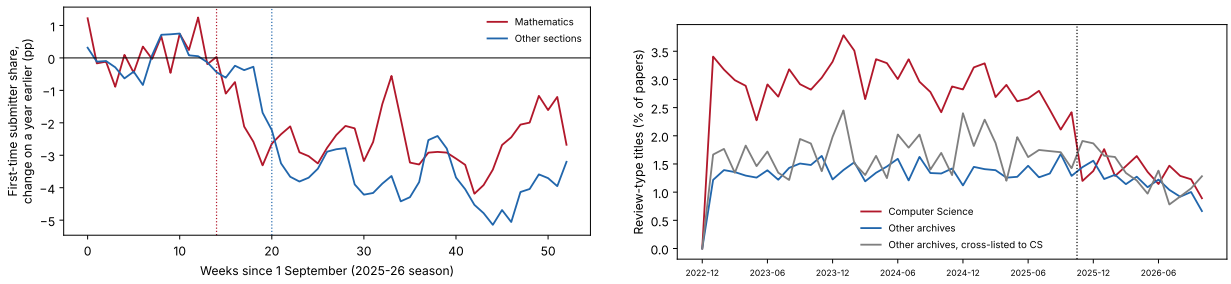
8.3 Catch-up cost: review articles in computer science

Review articles summarise a literature and lower the cost of catching up on it (McMahan and McFarland, 2021). On 31 October 2025, arXiv began to require that review and position papers in computer science be accepted after peer review before posting (arXiv, 2025a). The share of computer-science papers with review-type titles fell from 2.7% to 1.3%, from about 290 to 187 a month. Relative to other archives the fall is 1.32 percentage points (SE 0.10), and it survives a CS-specific trend. Reviews did not migrate to other primary categories cross-listed to computer science (Table 5, panel B). Within computer science, categories that had relied more on reviews lost more of them. A category with a 1 point higher pre-period review share lost 0.41 points more (SE 0.10), against an insignificant placebo a year earlier.

In the model, fewer reviews mean slower catch-up, a lower b_c and therefore lower mature density, more topics and fewer overshooting topics (Proposition 7(iv)). The short-run implication is less entry into existing topics where reviews mattered. The share of papers with a first-time author fell more in review-reliant categories: -0.70 per unit of exposure (SE 0.26). The same regression a year earlier gives -0.32 (SE 0.17), so part of the gradient predates the rule. We read the reduced form as suggestive. The topic-level predictions are stated in Section 11.

8.4 A cap on writing: two submissions per person per month

On 1 October 2026, arXiv limited every submitter to two new submissions per calendar month, without prior announcement and with no exemptions (arXiv, 2026b). The cap applies to the submitting account, not to the authors. Because shared names belong to different people, we measure excess submissions among names whose submissions in a half-month fall in a single archive. In the first half of each month of 2026 before the cap, 1.66% of papers were a submitter's third or later submission; in October the share was 0.38%, a residual consistent with shared names and proceedings accounts. The cap therefore removed about 1.3% of papers mechanically. Volume did not fall. Papers per weekday rose 0.220 log points from the first half of September to the first half of October 2026, against 0.175 on average in 2024–2025. Categories more exposed to heavy submitters did not grow less (Table 5, panel C). Per-submitter caps bind on a sliver of output and



(a) Endorsement rule

(b) CS review policy

Figure 4: Two arXiv rule changes. (a) First-time submitter share, change on the same week a year earlier (three-week moving average), Mathematics and other sections; dotted lines mark 10 December 2025 and 21 January 2026. (b) Share of papers with review-type titles by archive; the dotted line marks November 2025.

can be met by letting a co-author submit. In the model's terms, the cap is not a shock to writing speed, and the model predicts that a binding one, being a policy inside topics, would be largely dissipated by entry.

8.5 Drafting speed: author-level adoption of LLM-style writing

Proposition 7(i) predicts that faster drafting raises papers per researcher one for one. To measure drafting at the author level, we follow the excess-vocabulary literature (Liang et al., 2024; Kobak et al., 2025). We flag abstracts that use words whose frequency jumped after ChatGPT ("delve", "showcase", "underscore", "intricate", "pivotal", and similar). Their share rose from 3% of abstracts in 2022 to 11% in 2024. Among 68,467 submitters active in 2021–2022 and 2023, the 4,488 who used such words in at least half of their 2023 abstracts are adopters. Submitters who never used them are the comparison group. Classification uses 2023 papers only, so the outcome window, 2024–2026, is not conditioned on it. With submitter and field-by-half-year fixed effects, adopters' log(1 + papers per half-year) rose by 0.022 (SE 0.005), about 7% more papers, and by 0.024 (SE 0.007) in fields where AI is not the subject (Fig. 8 in Appendix D). Pre-period coefficients drift by up to 0.039, so part of this is a pre-existing trend. The drafting channel is present but small: adopters have so far used language models to write differently more than to write more.

8.6 What the shocks say about the model

The four designs do not test every proposition, and two of them are first stages whose outcomes are still accruing. Together they establish three things. The entry margin responds to its cost, which is what makes density respond little to volume. The supply of review articles, a lever on catch-up costs, responds to policy, and the first reduced form on entry has the sign the model predicts. And the detection of overlapping work does not run through passive visibility. Reading protects through deliberate search, which is why better search tools, not more prominent announcements, are the

Table 5: arXiv rule changes as shocks to entry, catch-up and writing

Design and outcome	Estimate	(SE)	Pre-period mean
A. Endorsement tightening (entry cost): Mathematics 10 Dec 2025, all sections 21 Jan 2026			
First-time submitter share, Mathematics, vs. year before	−0.024	(0.003)	0.141
First-time submitter share, other sections, vs. year before	−0.034	(0.003)	0.222
First-time submitter share, Mathematics vs. others, 10 Dec–20 Jan	−0.017	(0.004)	
Submitters with no prior arXiv authorship, Mathematics	−0.009	(0.003)	0.043
Submitters with no prior arXiv authorship, other sections	−0.009	(0.002)	0.054
log number of first-time submitters, other sections	−0.154	(0.035)	
B. Peer review required for CS review and position papers (catch-up cost), 31 Oct 2025			
Review-type share of papers, CS vs. other archives	−0.013	(0.001)	0.027
with a CS-specific linear trend	−0.010	(0.002)	
leakage: non-CS primary cross-listed to CS	−0.002	(0.001)	
Within CS: change in review share on pre-period review share	−0.41	(0.10)	0.027
placebo, one year earlier	−0.13	(0.09)	
Within CS: change in first-time-author share on review share	−0.70	(0.26)	
placebo, one year earlier	−0.32	(0.17)	
C. Two submissions per submitter per month (writing cap), 1 Oct 2026			
Third-or-later submissions in days 1–15, single-field names: Jan–Sep 2026	1.66%		
October 2026 (residual: shared names and proceedings accounts)	0.38%		
log papers per weekday, Oct 1–15 vs. Sep 1–15: 2026	+0.220		
same change in 2024 and 2025	+0.173, +0.178		
Category dose-response: change on exposure (vs. 2025)	+0.03	(2.95)	0.016
D. Author-level adoption of LLM-style writing (drafting speed)			
log(1 + papers per half-year), adopters vs. others, 2024–26, all fields	+0.022	(0.005)	0.48
fields where AI is not the subject (no cs, stat, eess)	+0.024	(0.007)	0.52

Notes: Panel A: weekly cells by announcement week, September 2024–October 2026; each week is differenced against the same week a year earlier; Newey–West standard errors (four lags). Panel B: monthly cells, 2023–2026, archive and month fixed effects, robust standard errors; dose-response across 29 CS categories with at least 300 papers a year, weighted by size, Nov–Oct years. Panel C: days 1–15 of each month, papers announced in their regular batch; exposure is the category's Jan–Sep 2026 share of excess submissions. Panel D: submitter and field-by-half-year fixed effects, standard errors clustered by submitter. Identities are name keys throughout (Appendix D).

channel through which AI could move the aware share. A per-person writing cap, finally, is not a lever on volume at all.

9 Quantification

This section puts numbers on the mechanisms. The aim is not to fit every moment of the arXiv record. It is to check that one parameterisation of the model reproduces the main facts with plausible primitives, and to read off its implications for rents, the speed limit, the incidence of research tools, topic life cycles and growth.

9.1 Strategy

We use the functional forms of Section 2: harmonic learning (2), exponential detection and the exact catch-up time (7). Time is measured in years. Table 6 lists the parameters. Some are set externally. The unit of density is fixed by $b = 100$ papers read per year at full-time reading, and production speed by $a = 4$ projects per year at full-time production. Careers leave a topic at $\rho = 0.10$ (ten-year tenure), the stock's relevance decays at $\delta = 0.25$ (a half-life of about three years), topics become obsolete at $\xi = 0.04$, and a newborn topic produces $P_0 = 2$ papers a year. The credit block comes from Fact 3 through Lemma 1. With exponential follow-on credit and a pivot that costs 30% of a project, the +29% premium of aware follow-ons and the −19% penalty of unaware duplicates imply that 31% of detected overlaps proceed and that detection averts $\Delta/\lambda = 37\%$ of erosion. We set concurrent overlap to zero: our moments do not separate it from erosion by unread earlier work, and attributing all erosion to unread work gives reading its largest protective value. Explorers found a viable topic at rate $s = 0.14$, so that half of them found one before their spell ends ($\omega = 1/2$). The frontier block splits the calibrated frontier value into exploration and founding; it does not affect the cross-section, and we report results for ω between 0 and 3/4.

The remaining parameters are calibrated jointly: learning (β, γ), erosion $(\theta\lambda)$, search precision (σ), the frontier value F, catch-up speed b_c, and the distributions of diffusion rates and fertility. The targets are the corrected returns profile of Fact 2: the gradient per standard deviation of density, the returns of the sparsest, fourth and densest fifths relative to the middle one (−11%, +2%, −2%), and the 12% of researcher-years beyond the peak. We add the recall-corrected aware share of 60% at mature density (Fact 4), which disciplines σ, and the share of incumbents' time spent reading. Faculty report about 144 hours a year reading scholarly articles (Tenopir and King, 2008), about 7% of a working year. This target disciplines the time-share value of reading tools (Lemma 5), which the previous version of the model left free. A soft target puts the incumbents' log premium $(\rho + \xi)T$ at about 0.15, about a year of catch-up per seven-year spell in a topic. Since the measured gradient also reflects audience, this is generous to the catch-up channel. We simulate the stationary cross-section of topic life cycles and run the arXiv regressions on it. As in the data, density is measured with error. Only 58% of a paper's nearest neighbours fall in its own fine area. We add classical noise with a standard deviation of 0.3 in log density, which implies a reliability

Table 6: Parameters

Parameter	Meaning	Value	Source
Set externally			
b	papers read per year at full-time reading	100	normalisation (unit of P)
a	projects per year at full-time production	4	normalisation
λ	credit lost by an unaware duplicate	0.19	Fact 3 (-19%)
κ_p	credit lost by pivoting	0.30	assumption
Δ/λ, q	detection value; share proceeding	0.37, 0.31	Fact 3 ($+29\%$), Lemma 1
δ	decay of the stock's relevance	0.25	half-life ≈ 2.8 years
ρ	rate of leaving a topic	0.10	ten-year topic tenure
ξ	topic obsolescence	0.04	25-year topic life
P_0	flow of a newborn topic	2	a few papers per year
s	explorers' success rate	0.14	$\omega = s/(\rho+\xi+s) = 0.50$
Calibrated (Table 7)			
β, γ	learning efficiency, curvature	39.47, 0.157	hump in returns
$\theta\lambda$	erosion per earlier paper	0.0066	fall at the top
F	frontier value	1.760	sparsest fifth
b_c	catch-up reading speed (speed limit)	38.6	premium, share beyond peak
σ	search precision	1.61	recall-corrected aware share
g_{med}, s_g	diffusion rate: median, log-sd	0.180, 0.29	share beyond peak, fifths
s_ζ	log-sd of topic fertility	0.043	gradient across topics
e	explorers' output (life cycle)	1.696	$F = (1-\omega)e + \omega\bar{U}_F$
η_F	frontier depletion	0.0166	$\mathcal{E} = 0.15$ (Fact 1), (13)

of about 0.8. Researchers who move between topics supply the identifying variation in an author-fixed-effects design, and the model has no ability differences. The baseline analogue is therefore the cross-section of researcher-years. The within-topic analogue, which uses only life-cycle variation, is a robustness check. The search is a quasi-random scan of 2,048 points followed by Nelder–Mead from the best points and from the previous calibration, with the winner selected on a fine simulation grid (Appendix C). The frontier's depletion η_F is then set from (14) with $\mathcal{E} = 0.15$ (Fact 1).

9.2 Fit

Table 7 and Fig. 5 compare model and data. The baseline calibration reproduces the aware share at mature density (61% against 60%), the reading share of incumbents' time (7.8% against 7%), most of the share of researcher-years beyond the peak of the payoff curve (10% against 12%) and most of the rise from the sparsest fifth (-8.6% against -11%). Without targeting it, the peak lies 0.70 standard deviations above mean density, against about one in the data. The gradient falls short (5.2% against 6.7%), and so does the incumbents' premium, which is generous to the catch-up channel because the measured gradient also reflects audience. Exact catch-up makes the problem harder than it was with the linear approximation: the newcomer's payoff curve now falls about twice as steeply at the mature density, and fast topics stop well below the densities they reached before, so the life cycle generates less overshooting for a given distribution of diffusion rates.

The model still does not reproduce the top of the profile. In the corrected data the densest fifth earns *less* than the middle (-2%). In the model the fourth and densest fifths earn $+1.3\%$ and

Table 7: Fit to the arXiv facts

Moment	Data	Baseline	Within-topic
Gradient per SD of density	0.067	0.052	0.058
Sparsest fifth vs. middle	−0.110	−0.086	−0.110
Fourth fifth vs. middle	0.020	0.013	0.009
Densest fifth vs. middle	−0.020	0.022	0.013
Researcher-years beyond the peak	0.120	0.098	0.108
Incumbents' log premium $(\rho+\xi)T$	0.150	0.103	0.159
Aware share of flagged overlap	0.60	0.61	0.60
Reading share of incumbents' time	0.070	0.078	0.087
Peak of v (SD above mean)†	1.00	0.70	0.93
Quadratic: linear term†	0.094	0.038	0.050
Quadratic: squared term†	−0.047	−0.007	−0.012

† Not targeted. Baseline: cross-section of researcher-years (movers across topics). Within-topic: topic fixed effects, recalibrated. Density is the researcher-weighted standardised log flow, measured with noise as in the data. The reading share is from Tenopir and King (2008).

+2.2%. The source of the miss is composition. The densest researcher-years in the model include the most fertile mature topics, whose incumbents earn high returns, alongside the overshooting topics that pull returns down. Erosion in mature topics is small, about 2.9% of credit, so the payoff curve falls only gently beyond its peak. The fall at the top points to crowding costs in dense topics beyond duplication and catch-up, to concurrent overlap, which we set to zero, or to audience effects that flatter the middle of the distribution. We regard it as the model's main quantitative failure. 56% of researcher-years are in mature topics, 20% in topics consolidating after a growth spurt or declining after a stall, and 24% in growing topics. The within-topic calibration fits its moments more closely (Table 7). Its median diffusion rate is low, 5% a year, so young topics grow slowly and the life-cycle variation it relies on is spread over many years; relaxing the bounds of the search does not change it.

9.3 What the calibrated model says

Table 8 collects the implications.

Rents and the speed limit. The payoff curve peaks at $\hat{P} = 12$ papers a year, the newcomer's payoff curve at $\hat{P}_w = 4.3$, and mature topics sit between them, at $P^m = 6.5$. The speed limit is $b_c = 39$ papers a year: mature topics sit at 17% of it, and the most fertile at 33%. Catching up takes 0.7 years at the mature density and 1.5 at the peak. Incumbents earn a premium of 11% over the frontier, and the cross-topic returns gradient, the catch-up elasticity $\varepsilon_\Gamma = 0.113$, is steeper than the premium, as Proposition 5(iv) requires. Even with rising incumbent returns ($\varepsilon_v(P^m) = 0.048 > 0$), the newcomer's payoff curve is falling at the mature density ($|\varepsilon_w| = 0.065$), so mature topics are too crowded once the catch-up burden is counted (Corollary 1). Moving every topic to the peak of w raises output per researcher by 1.2% (3.2% in the within-topic calibration). The frictionless benchmark, in which catch-up is free and topics fill to the density where $v = F$, would dissipate

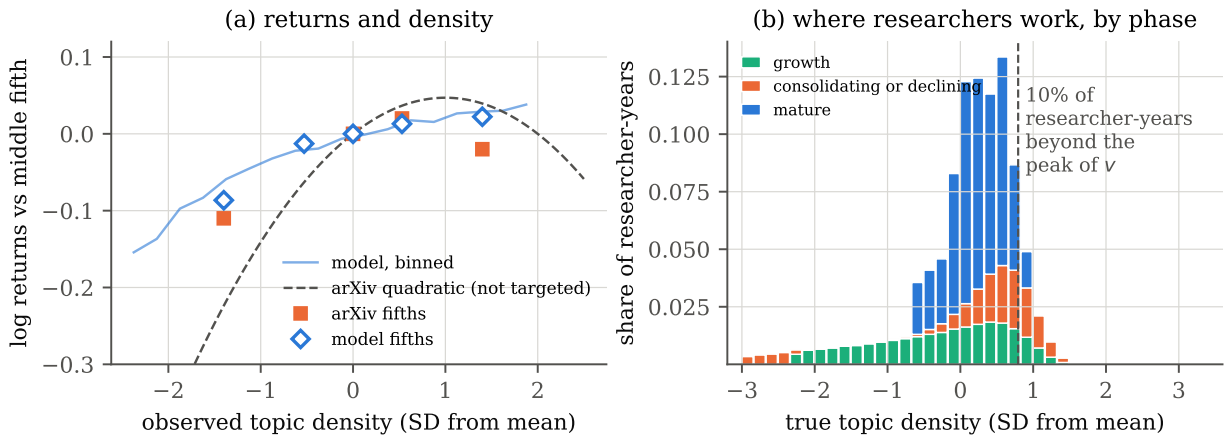


Figure 5: Returns and density in the model. Left: log returns by observed (noisily measured) topic density, relative to the middle fifth, in the calibrated cross-section, with the arXiv fifths and the arXiv quadratic (not targeted). Right: the distribution of researcher-years by true density and life-cycle phase. Researcher-years beyond the peak of v are concentrated in topics consolidating after fast growth.

11% of output per researcher. That number describes a counterfactual economy, not the calibrated one.

The incidence of research tools. Tables 2 and 9 and Fig. 6 report the effects of 30% improvements in each channel. Reading takes 7.3% of an incumbent's time at the mature density, so faster reading is worth $r^*/(1-r^*) = 0.078$ per log point at a fixed topic: a 30% faster reader gains 1.96 log points, a 30% faster producer 26.2. Per hour saved, the two are worth the same. Entry then ranks them. The producer keeps almost everything, 26.0 log points. If explorers never stay in the topics they open, the reader keeps 0.27 log points, 14% of her gain, close to the floor $\mathcal{E}D$ of (18). If founders stay with probability one half, she keeps 1.46, or 74%. Per hour saved, a production tool is therefore worth between 1.3 and 7.2 times a reading tool. Catch-up tools keep 16% without founders and 40% with them; founders never catch up, so their gain reaches the frontier only through the density of the topics founders end up in.

On the simulated life cycle the founders' channel is weaker for catch-up tools and somewhat weaker for reading tools. Founders spend much of their tenure in topics still growing at a pace set by diffusion, so the frontier value rises by 42% of a reading gain and by only 1% of a catch-up gain. Output per researcher-year, which also counts the rents of newcomers who enter growing topics before entry pays only the frontier value, rises by more: 63% of a reading gain (40% without founders), 17% of a catch-up gain and 99% of a production gain. These rents raise life-cycle output per researcher 2.5% above the frontier value. The footprints in Table 2 follow Proposition 7. Faster reading raises density by 7.8 log points; faster catch-up raises it by 26.3 and lowers the overshooting threshold by 13.1 percentage points, so more topics grow too fast; better search raises the aware share by 12.9 percentage points.

Table 8: Rents, the speed limit, welfare and growth in the calibrated model

Object	Value
Payoff curve and catch-up	
Peak of the payoff curve $\hat{P}$ (papers per year)	11.8
Mature density P^m; speed limit b_c	6.5; 38.6
Catch-up time at P^m / at $\hat{P}$ (years)	0.74 / 1.45
Incumbents' premium over the frontier, $\Gamma(P^m)^{-1} - 1$	10.9%
Cross-topic returns gradient $\varepsilon_\Gamma(P^m)$	0.113
Slope of the entrant payoff curve $\lvert\varepsilon_w(P^m)\rvert$	0.065
Newborn margin $\ln[v(P_0)/F]$	1.28%
Welfare and policy	
Frictionless benchmark: rent dissipated, $1 - F/\hat{v}$	11.2%
Stationary planner's gain, $\hat{w}/F - 1$	1.2%
Planner's gain if all overlap losses are transfers ($\tau = 1$)	0.4%
Life cycle: gain from a flat frontier subsidy s^o	0.13%
Life cycle: gain from an entry tax proportional to the stock	0.72%
Topic life cycles	
Take-off threshold g_0 (per year)	0.022
Overshooting threshold g^*, myopic newcomers	0.205
Closed-form bound at the peak, g^*_∞	0.214
Overshooting threshold, forward-looking newcomers	0.240
Sufficient bound, forward-looking	0.291
Growth	
Short-run bias in $\mathcal{E}$ from topic-age composition (7 years)	−0.064
Frontier depletion η_F (from $\mathcal{E} = 0.15$)	0.0166
Stepping on toes, topics adjust: $1 - \mathcal{E}\lvert\varepsilon_w\rvert$	0.990
Stepping on toes, open topics fixed: $1 + \varepsilon_w/D$	0.941

Readership. Valued by readership ($\alpha = 1$ in Proposition 8), a 30% faster pen adds 1.3 log points and 30% faster reading 21.2: a faster pen leaves total reading unchanged and cuts reads per paper by 25%. With readership given half the weight of credit ($\alpha = 1/2$) the two tools are worth about the same, 13.6 and 11.3 log points. Which tool scales science depends on the unit, and the unit is a choice about what science values.

Topic life cycles. A newborn topic beats the frontier by 1.28%, so its first followers join, but a stationary topic of the same density would not be worth joining. Topics that grow more slowly than 2.2% a year stall in the valley and are abandoned, and topics that grow faster than 20.5% overshoot the peak (Proposition 9); with foresight the overshooting threshold is 24.0%. These are fast but not unusual rates for hot topics on arXiv.

Growth. Inverting (14) with $\mathcal{E} = 0.15$ gives a frontier depletion of $\eta_F = 0.017$ (0.012 without founders), and the stepping-on-toes parameter is $\lambda_J^{\text{eq}} = 1 - \mathcal{E}|\varepsilon_w| = 0.990$ when topics adjust and $\lambda_J^M = 1 + \varepsilon_w/D = 0.941$ when open topics are held fixed (Proposition 10). Over the arXiv range $\mathcal{E} \in [0, 0.27]$, λ_J^{eq} lies between 0.982 and one. The short-run bias from topic-age composition (Remark 4) is −0.064 over a seven-year window, so even the upper end of the long-run range keeps

Table 9: How much of each tool survives entry, and readership

	Faster reading		Faster catch-up		Faster production	
	formula	life cycle	formula	life cycle	formula	life cycle
A. Share of the gain at fixed topics that survives entry (%)						
$\omega = 0.00$	17	40	17	11	99	99
$\omega = 0.25$	50	44	31	17	99	99
$\omega = 0.50$	74	63	41	17	99	99
$\omega = 0.75$	91	79	48	16	99	99
B. Long-run gain valued by readership, $\omega = 0.5$ *(log points* $\times 100$*)*						
$\alpha = 0.0$	+1.5		+1.0		+26.0	
$\alpha = 0.5$	+11.3		+6.3		+13.6	
$\alpha = 1.0$	+21.2		+11.5		+1.3	

Panel A: *formula* is the frontier-incidence pass-through at the mature density (Theorem 3) with founders' topics valued at maturity and η_F set from the volume elasticity; *life cycle* re-solves the founders' fixed point $F = (1-\omega)e + \omega \bar{U}_F(F)$ on the simulated life cycle (no depletion) and measures output per researcher-year with exploration, catch-up and diffusion rents counted. Panel B: output valued as $A^\alpha Y$, where $A = (b/a)r^*/(1-r^*)$ is reads per paper; $\alpha = 0$ is credited output, $\alpha = 1$ readership.

λ_J^{eq} close to one. Topic crowding and access depletion explain little of the diminishing returns to research found by Bloom et al. (2020). Their estimates require λ_J well below one, which in our framework must come from fertility depletion (Appendix B.1) or from selection.

10 Implications: topics that grow too fast, growth and policy

10.1 When innovation is too fast

The stock of literature lags its flow. A topic whose literature grows fast has a small stock relative to its current density, so joining it is cheap. This subsection shows that such topics overshoot their best density, that topics that grow too slowly never take off, and that overshooting does not rely on newcomers being myopic.

Consider a topic born with flow P_0 and an empty stock. Potential newcomers learn about the topic over time. While entry pays, they join at a rate that makes the topic's flow grow at the diffusion rate g, $P(t) = P_0 e^{gt}$, and the stock follows $\dot{S} = P - \delta S$ from $S(0) = 0$, so that along the growth path

$$S(P) = \frac{P}{g+\delta}\Big[1 - \Big(\frac{P_0}{P}\Big)^{(g+\delta)/g}\Big] < \frac{P}{g+\delta}. \tag{19}$$

Diffusion is the only friction on the speed of entry. Entry stops at the first date t_s at which the value of joining falls to the frontier value, and $P_s = P(t_s)$ is the *stopping density*. After t_s no one joins, members leave at rate ρ, and the topic thins. If it stopped above its mature density, entry resumes when it has thinned to P^m. (We use the stationary stock for that date.) If it stopped below, it stalled in the valley described next.

Two kinds of newcomer bracket the possibilities. A *myopic* newcomer expects the flow to stay at its current level, so her value relative to the frontier is $v(P)\big[1 + \delta S/(b_c - P)\big]^{-\nu}/F$ (Lemma 2).

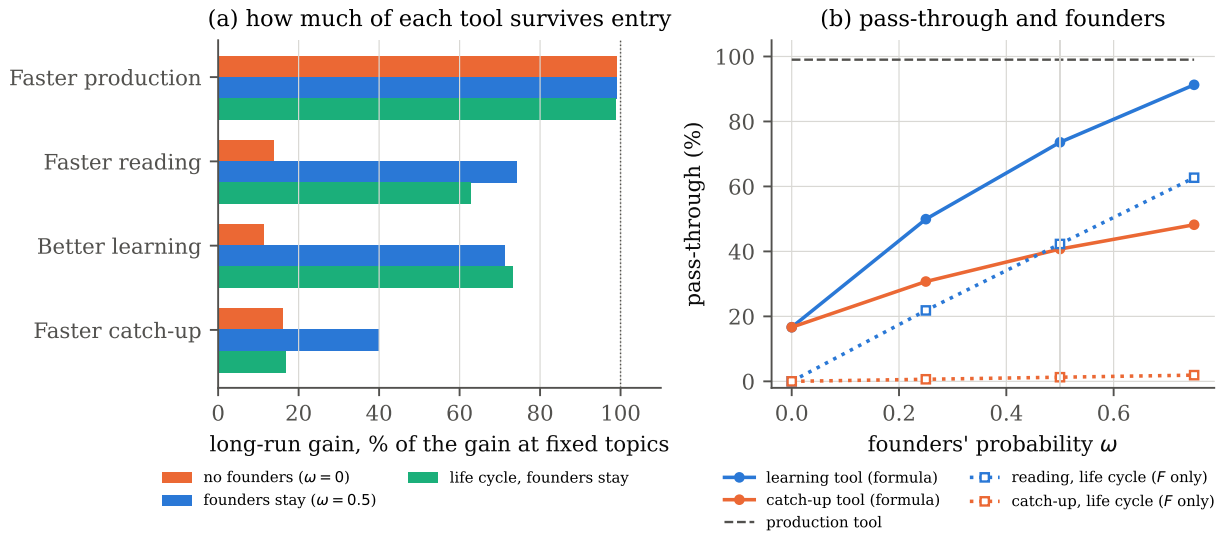


Figure 6: The incidence of research tools in the calibrated model (30% shocks). Left: long-run gain in output per researcher as a share of the gain at fixed topics, without founders, with founders who stay (stationary), and on the simulated life cycle. Right: the share that survives entry as a function of the founders' probability ω, from the incidence formula and, for the frontier value alone, on the life cycle.

A *forward-looking* newcomer correctly anticipates that the topic will thin once entry stops. She reads her backlog (6) along the actual path of the flow and counts payoffs only once it is cleared:

$$U(P_s, S_s) = (\rho + \xi) \int_{T(P_s,S_s)}^{\infty} e^{-(\rho+\xi)t}\, v\big(\max\{P_s e^{-\rho t}, P^m\}\big)\, \mathrm{d}t,$$

where $T(P_s, S_s)$ is now her catch-up time when the flow thins. Her value is U/F.

Proposition 9 (Too slow and too fast)**.** *Let mature topics lie on the rising side of v, with $\hat{P} < b_c$, and let the first follower of a newborn topic beat the frontier, $v(P_0) > F$.*

(i) (Take-off.) *Under myopic entry the stopping density $P_c(g)$ is nondecreasing in g and below b_c for every g. If $w(P_0) < F$, there is a take-off threshold $g_0 > 0$: topics that grow more slowly stall below P^m, in the valley where a stationary topic is not worth joining; topics that grow faster reach at least P^m.*

(ii) (Overshooting.) *A topic overshoots the peak, $P_c(g) > \hat{P}$, if and only if $g > g^* \equiv \max\{g_0, g_{\hat{P}}\}$, where $g_{\hat{P}}$ is the rate at which a myopic newcomer at the peak density still beats the frontier, $\hat{v}\,[1 + \delta S(\hat{P})/(b_c - \hat{P})]^{-\nu} = F$. It satisfies*

$$g_{\hat{P}} \le g^*_\infty \equiv \frac{\delta\, \hat{P}}{(b_c - \hat{P})\,[(\hat{v}/F)^{1/\nu} - 1]} - \delta, \tag{20}$$

with equality in the limit of a negligible birth flow, $P_0/\hat{P} \to 0$, in the stock.

(iii) (Forward-looking entry.) *A topic that takes off and grows faster than*

$$g^*_{RE} \equiv \delta\Big[\frac{\hat{P}\,(b_c - P^m)}{P^m\,(b_c - \hat{P})} - 1\Big]$$

overshoots the peak under forward-looking entry.

(iv) (Consolidation.) *After entry stops above P^m, incumbents stay, since their continuation value exceeds F. The topic thins at rate ρ to P^m and spends $\ln(P_s/\hat{P})/\rho$ periods beyond the peak.*

The proposition gives a clean answer to the question whether "too fast" is an equilibrium phenomenon. The friction that produces overshooting is the stock lag, not myopia. Part (iii) has a simple logic. If entry stopped before the peak, the future path of the topic would lie between P^m and the peak, where v is at least $v(P^m)$, and the flow would fall, so a newcomer would read less than if it stayed put. A newcomer at the stopping date faces a stock of at most $P/(g+\delta)$. Whenever g exceeds g^*_{RE} her catch-up is shorter than in a mature topic, so she earns more than a newcomer to a mature topic, who earns exactly F, and entry could not have stopped.

Part (i) is new to this version and follows from the exact catch-up time. A newborn topic is worth joining because it has no stock. If it grows slowly, its stock catches up with its flow before the topic is dense enough to be worth joining in its own right, and entry stalls. In the calibration a stalled topic is never worth joining again: its flow falls faster than its stock, so it thins and is abandoned. Topics that grow too slowly die in the valley between their birth and maturity; topics that grow too fast overshoot. Both are failures of the same coordination, in opposite directions. In the calibration the take-off threshold is 2.2% a year, the myopic overshooting threshold is 20.5%, against a closed-form bound of 21.4%, and forward-looking entry overshoots above 24.0%, with a sufficient bound of 29.1%. A topic that grows at 200% a year stops at 26 papers a year with foresight, against 24 without (Fig. 7), and below the speed limit of 39 in both cases.

The result locates the overcrowded researcher-years in topics whose literature grew fast, which persist long after the rush because incumbents never pay the catch-up cost that deters newcomers. It also identifies who sets the speed limit. The thresholds fall with catch-up speed b_c, so review articles and literature-summarising tools that make catching up cheap make overshooting *more* common.

10.2 Stepping on toes, measured

Growth models since Jones (1995) let the flow of new ideas scale with researchers as N^{λ_J}, $\lambda_J < 1$, and call $1-\lambda_J$ "stepping on toes". The parameter is reduced form, and what it holds fixed matters. Here the flow of credited output is $Y = N\,a\,F$ (Lemma 4).

Proposition 10 (Stepping on toes)**.** *In the large-topic limit,*

$$\lambda_J^{\text{eq}} \equiv \frac{\mathrm{d}\ln Y}{\mathrm{d}\ln N} = 1 - \mathcal{E}\,|\varepsilon_w(P^*)|, \qquad \lambda_J^{M} \equiv \frac{\partial \ln Y}{\partial \ln N}\Big|_M = 1 + \frac{\varepsilon_w(P^*)}{D}. \tag{21}$$

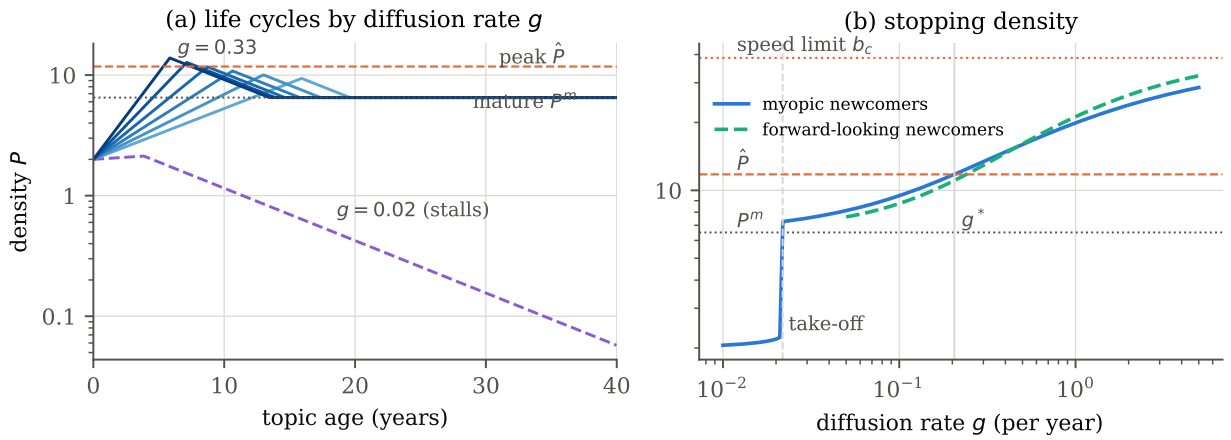


Figure 7: Topic life cycles. Left: density paths for topics with different diffusion rates g in the calibrated model. A topic that grows too slowly stalls and is abandoned (dashed); topics that grow fast overshoot the peak $\hat{P}$ and consolidate slowly. Right: the stopping density as a function of g for myopic and forward-looking newcomers. It jumps at the take-off threshold and stays below the speed limit b_c.

λ_J^{eq} lets the number of topics adjust; λ_J^M holds the open topics fixed and adds researchers to them. If, in addition, topics opened later are less fertile, with fertility elasticity $\eta_f \geq 0$ (Appendix B.1), the first expression is an upper bound.

Jones's parameter holds the stock of knowledge fixed. If the open topics count as knowledge, the relevant object is λ_J^M, the return to adding researchers to the topics that exist, which the slope of the newcomer's payoff curve alone determines. If topics are a choice, the relevant object is λ_J^{eq}, which is built from two observables: the volume elasticity of density, $\mathcal{E}$ (Fact 1), and the slope of the newcomer's payoff curve at equilibrium density, ε_w. Crowding does not amplify frontier depletion. It buffers it. The density margin absorbs part of the growth in researchers, so fewer new topics are opened and the frontier deteriorates more slowly. In the calibration $\lambda_J^{\text{eq}} = 0.990$ and $\lambda_J^M = 0.941$. With $\mathcal{E}$ between 0 and 0.27, λ_J^{eq} lies between 0.982 and one (Section 9). On either definition, topic crowding and access depletion explain little of the measured decline in research productivity (Bloom et al., 2020). The rest must come from falling fertility of new topics (η_f, which density data do not identify) or from selection on researcher ability (Ekerdt and Wu, 2024).

10.3 Policy

The policy implications follow from where the rents go. Anything that raises the payoff of working inside existing topics is competed away unless it also reaches the people who open new ones. Anything that raises the frontier is not. In the stationary economy the planner puts each topic at the peak of the newcomer's payoff curve, and a flat frontier subsidy decentralises the optimum (Proposition 4). Topics are not born at their mature density, though, and the subsidy has a side effect at birth.

Proposition 11 (Policy that depends on a topic's age)**.** *Consider the life-cycle economy at a given frontier value F.*

(i) A flat subsidy s to exploring, or a flat tax s on joining open topics, deters the first follower of every newborn topic if $s > v(P_0) - F$. No topic then takes off.

(ii) An entry tax proportional to the stock a newcomer must read, τS, is zero in a newborn topic and never deters its first followers. With $\tau^o = \delta(\hat{w} - F)/\hat{P}_w$ it puts mature topics at the peak of the newcomer's payoff curve, as in the stationary optimum.

In the calibration the stationary subsidy, 1.2% of the frontier value, is almost as large as the newborn margin, $\ln[v(P_0)/F] = 1.28\%$, so a flat subsidy would bring the economy to the edge of the regime in which new topics attract no followers. Even short of that edge it is a blunt instrument. It raises the bar for every newcomer, including those who carry a young topic through the valley before it is worth joining in its own right, so the take-off threshold rises from 2.2% to 3.6% and more young topics die. With founders who stay, the frontier value itself changes by -0.66%, and output per researcher-year on the life cycle rises by only 0.13%. The stock-proportional tax is zero at birth and small while a topic's stock is still small, so it raises the take-off threshold by less, to 3.1%; it brings mature topics close to the peak of the newcomer's payoff curve and raises output by 0.72%, more than half of the stationary planner's gain. In practice a tax on the stock means credit, grants and prizes that reward contributions to young topics and to new lines of work more than contributions to old, crowded ones. Faster catch-up tools work in the opposite direction: they lower the effective tax on joining crowded topics.

11 Out-of-sample predictions

The catch-up extension was added after the returns-by-density evidence was seen, so that evidence cannot test it. The predictions below were not used to build the model. They are stated before the relevant data were analysed, and several before the data exist. To make the claim verifiable they should be deposited in a public registry before they are taken to the data. The replication package contains the text.

Predictions testable with existing data.

P1. *Overcrowding is young and fast.* The fall in returns beyond the peak should be concentrated in clusters whose flow grew faster than their stock, that is, with a high flow-to-stock ratio P/S (Proposition 9). Among mature clusters the returns curve should be monotone.

P2. *Slow clusters die young.* Among clusters born in the same year, those whose flow grew slowly in their first years should be abandoned more often, and the hazard should fall sharply above a threshold growth rate (Proposition 9(i)).

P3. *Returns follow the stock, density follows the flow.* Holding flow density fixed, returns should

rise with the cluster's depreciated stock of literature (Proposition 5(ii): the premium is $(\rho + \xi)T(P, S)$).

P4. *Newcomers start slowly where stocks are large and flows are fast.* The lag between a newcomer's arrival in a cluster and her first paper there, and the length of her first reference list, should rise with the cluster's stock, and convexly with its flow as the flow approaches what newcomers can read (Lemma 2). Incumbents should show no such gradient.

P5. *The cross-topic gradient is the catch-up elasticity plus audience.* The returns gradient from researchers who switch clusters should exceed the log premium of incumbents over pioneers in the clusters they join, by the gap between the catch-up elasticity and the premium (Proposition 5(iv)) and by the audience effect, which can be measured with citations from outside the cluster.

P6. *Close overlap is pinned; broad neighbourhoods fill.* The volume elasticity should rise with the similarity radius, from near zero at the overlap radius toward the fixed-field benchmark at broad radii (Fact 1 already shows this ordering), and the ordering should survive embedding-based similarity. The ratio of the two elasticities measures how easily topics open inside existing fields (Proposition 13).

P7. *Founders stay.* The share of a cluster's founding authors who keep publishing in it, and their returns there relative to their earlier exploratory work, measure the founders' share of the frontier's value, θ_F. Areas with a larger θ_F should gain more from reading tools (Theorem 3).

P8. *Stepping on toes across areas.* Areas with a larger volume elasticity of density should show stronger diminishing returns to researchers in citation output, with $1 - \lambda_J^{\text{eq}} = \mathcal{E}|\varepsilon_w|$ area by area (Proposition 10).

P9. *Search costs and detection.* Google was blocked in mainland China from 2014 (Zheng and Wang, 2020). Relative to Hong Kong-based and overseas Chinese authors in the same topics, mainland-based authors' close overlaps should have become less often aware, and their papers more often unaware duplicates, after 2014 (Lemma 1 with lower σ). The share of covering papers they pivot away from cannot be observed, so the prediction is on the aware share and on the citation penalty of their overlaps.

Predictions for the arXiv rule changes of 2025–2026.

P10. *Endorsement.* Relative to the other sections, Mathematics in 2026 and all sections in 2026–2027 relative to trend should show fewer pioneer papers (new clusters) in proportion to the fall in entrants. Local density at the overlap radius, the flagged overlap rate and the aware share should not change detectably (Proposition 7(ii) with small $\mathcal{E}$).

P11. *Review articles.* In computer-science clusters where reviews had been common, mature density should fall and the number of clusters rise relative to other computer-science clusters over

2026–2027. Fast-growing clusters there should overshoot less (higher g^*) and newcomers' first papers should come later and with longer reference lists (Proposition 7(iv), P3).

P12. *Submission cap.* The cap should leave total volume, papers per author and the overlap rate within a few percent of trend. Submissions by previously heavy submitters should be replaced partly by submissions in their co-authors' names.

P13. *Reading tools.* When literature-summarising tools are widely used, high-uptake fields where AI is not the subject should form fewer new clusters, thicken existing ones, and show more overshooting by fast-growing clusters. Overlap measured by meaning should rise in dense clusters, and citations per author-year should rise by much less than the time the tools save, except where the people who open new clusters use them (Proposition 7(iii)–(iv), Theorem 3).

P14. *Production shocks and closure.* A large, fast rise in production speed should raise the flow of the densest clusters most, lengthen newcomers' time to first paper there, and cut entry into them for a while (Theorem 2(iv)).

P15. *Better search raises the aware share.* If search tools improve, the share of close overlaps that cite the earlier paper should rise, with no offsetting fall in the overlap rate in sparse clusters (Proposition 7(v)).

Two earlier predictions have already been settled. Equal returns (full dissipation) is rejected, and the catch-up cost replaces it. A detection channel that runs through the prominence of a paper's announcement is rejected by the listing-rank design (Section 8), which locates detection in deliberate search.

12 Conclusion

Is innovation too fast? The model gives the question two precise meanings. The first is a speed limit. A literature can grow faster than the people outside it can read, and when it does, newcomers cannot join it, however rewarding it is for those inside. Near the limit, rents rise steeply; a sudden rise in production speed can push the densest topics past it; and because the stock of literature lags its flow, topics that grow fast overshoot their best density even when newcomers see it coming, while topics that grow too slowly never take off. The second meaning is the one the debate usually has in mind: whether papers arrive faster than science can absorb them. Here the answer is that researchers do not pile into the same topics until duplication explodes. They spread out, and the duplication rate stays where entry pins it. The arXiv record bears this out. Volume more than doubled while close overlap barely moved, and the entry margin responds to its cost.

The policy-relevant result is an incidence principle. In the long run, output per researcher equals the value of opening a new topic, so a research tool is worth what it adds to that value, not what it adds to the topics that already exist. Inside a topic, every tool is worth the time it saves. In equilibrium, tools that speed up production keep almost all of their value; tools that speed up reading keep the part that reaches founders who stay in the topics they open; tools that speed

up catching up keep the least. The asymmetry is no longer an assumption about pioneers but a measurable object: the share of the frontier's value that founders earn as incumbents, bounded below by the volume elasticity of density. Valued by readership rather than by credited output, the ranking reverses, which is a reminder that "scaling science" is a statement about units.

Several limits deserve emphasis. Credit is measured by citations, which also reward audience, so the rising arm of the returns profile is an upper bound on the catch-up channel. The founders' share is not yet measured, and the pass-through of reading tools depends on it. The causal test of the detection channel is a bounded null for passive visibility. It points to search costs as the right shock, and we have stated that design in advance rather than run it. The quasi-experiments on entry and catch-up costs are, so far, first stages with early reduced forms. And the idea space is coarse: topics are cells of fixed size, nested in fields. A continuous idea space in which topics form endogenously, with a wide learning kernel and a narrow covering kernel, is the natural next step, and it would turn the radius pattern of the volume elasticity from a fact into a test.

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A Proofs

Throughout, "concave" means concave on the stated convex domain, and ∂_R, ∂_P denote partial derivatives.

The example satisfies Assumptions 1 and 2. The function $h(R,P) = RP/(R+P)$ is half the harmonic mean of R and P. It is concave, nondecreasing and homogeneous of degree one on $\mathbb{R}^2_+$, with $h \le \min\{R,P\}$. So $\ln(1+\beta h)$ is concave as a nondecreasing concave transformation of a concave function, and Q is bounded by $(1+\beta R)^\gamma$ for fixed R. Supermodularity: $\partial_R \ln Q = \gamma\beta h_R/(1+\beta h)$ with $h_R = P^2/(R+P)^2$ and $h_{RP} = 2RP/(R+P)^3$, so

$$\partial_P \partial_R \ln Q = \gamma\beta\,\frac{h_{RP}(1+\beta h) - \beta h_R h_P}{(1+\beta h)^2} = \gamma\beta\,\frac{2RP(R+P) + \beta R^2 P^2}{(R+P)^4(1+\beta h)^2} \ge 0.$$

For $\varphi(c) = 1 - e^{-\sigma c}$: $\varphi(0) = 0$, $\varphi' = \sigma e^{-\sigma c} > 0$, $\varphi'' < 0$.

Proof of Lemma 1. Earlier covering papers form a Poisson count K with mean θP. Each is detected independently with probability φ, so detected and undetected counts K_d, K_u are independent Poisson with means $\theta P\varphi$ and $\theta P(1-\varphi)$. Concurrent covering projects form an independent Poisson count K_c with mean $\theta_c P$. Credit is the product of $(1-\lambda)$ over undetected papers, of independent factors with mean $1-\ell$ over detected papers, and of $(1-\lambda_c)$ over concurrent projects. Using $\mathbb{E}[t^K] = e^{-\mu(1-t)}$,

$$C = \mathbb{E}\big[(1-\lambda)^{K_u}\big]\,\mathbb{E}\big[(1-\ell)^{K_d}\big]\,\mathbb{E}\big[(1-\lambda_c)^{K_c}\big] = \exp\{-\theta P[\lambda - (\lambda-\ell)\varphi] - \theta_c\lambda_c P\}.$$

Flagged overlap consists of undetected covering papers (rate $\theta P(1-\varphi)$) and detected ones that proceed (rate $\theta P\varphi q$). The aware share is the ratio of the second rate to the sum. Proceeding happens iff $\delta > 1 - \kappa_p$, which gives the conditional mean. □

Proof of Lemma 2. With a constant flow, (6) is linear with rest point $B^\star = (P - b_c)/\delta$, so $B(t) = B^\star + (S - B^\star)e^{-\delta t}$. If $P < b_c$, $B^\star < 0$ and B reaches zero at $e^{-\delta T} = -B^\star/(S - B^\star) = (b_c - P)/(\delta S + b_c - P)$, which is (7). If $P > b_c$, $B(t)$ converges to $B^\star > 0$ from $S > 0$ and never reaches zero; if $P = b_c$, $B(t) = Se^{-\delta t} > 0$. With $S = P/\delta$, $1 + \delta S/(b_c - P) = b_c/(b_c - P)$, which gives the stationary T. A newcomer's spell in the topic ends at rate $\rho + \xi$ (her career ends or the topic dies). Her expected output over the spell divided by its expected length is $(\rho + \xi)\int_T^\infty e^{-(\rho+\xi)t} a\, v(P)\, \mathrm{d}t = a\, v(P) e^{-(\rho+\xi)T}$, and $e^{-(\rho+\xi)T} = (1 - P/b_c)^{(\rho+\xi)/\delta}$. In a stationary topic with entry flow ι, members who joined u periods ago are still there with probability $e^{-\rho u}$ while the topic lives, so the mass of members is ι/ρ and the mass that has caught up is $\iota e^{-\rho T}/\rho$. The producing share is $e^{-\rho T} = (1 - P/b_c)^{\rho/\delta}$, and (1) gives $P = a\, x^*(P)\, n\, (1 - P/b_c)^{\rho/\delta}$. □

Proof of Lemma 3. Value an option by the ratio of its expected output to its expected length (renewal reward). The exploring phase ends at rate $\rho + s$ (the career ends, or she founds a topic) and produces e per period. With probability $s/(\rho + s)$ it is followed by a founding phase that ends at rate $\rho + \xi$ and produces $\bar{U}_F/(\rho + \xi)$ in expectation, by the definition of $\bar{U}_F$. Expected output is $[e + s\,\bar{U}_F/(\rho + \xi)]/(\rho + s)$ and expected length is $[1 + s/(\rho + \xi)]/(\rho + s)$. Their ratio is $[(\rho + \xi)e + s\bar{U}_F]/(\rho + \xi + s) = (1 - \omega)e + \omega\bar{U}_F$. □

Proof of Lemma 4. Each researcher's career is a sequence of spells, each begun by a choice among options and ended by the end of her career, the death of her topic or, for explorers, the founding of a topic. By the renewal-reward theorem, her long-run output per period is the ratio of expected output to expected length of the spells she chooses, which is aF for every option she takes. In a stationary population of independent researchers the cross-section average of output per period equals this time average. □

Proof of Theorem 1. *(i)* Since $Q(0, P) = 1$ and $\varphi(0) = 0$, $v(P) \geq g(0, P) = e^{-(\theta\lambda + \theta_c\lambda_c)P}$. Because Q and φ are nondecreasing in R and $R \leq b$,

$$g(r, P) \leq \bar{Q}(b) \exp\{-\theta P[\lambda - \Delta\varphi(b/P)]\} \xrightarrow[P\to\infty]{} 0,$$

since $\varphi(b/P) \to 0$ and $\lambda > 0$. At $P = 0$, $g(r, 0) = 1 - r$, so $v(0) = 1$. Define $\ln C(R, 0) = 0$. Then $|\theta P\varphi(R/P)| \leq \theta P$ gives joint continuity of g on $[0, 1] \times \mathbb{R}_+$ ($g \to 0$ as $r \to 1$). Berge's maximum theorem gives continuity of v.

(ii) On $D = [0, 1) \times (0, \infty)$,

$$\ln g(r, P) = \ln(1 - r) + \ln Q(br, P) - (\theta\lambda + \theta_c\lambda_c)P + \theta\Delta\, P\varphi\left(\frac{br}{P}\right).$$

The first term is concave in r. The second is concave by Assumption 1 composed with a linear map. The third is linear. In the fourth, $(R, P) \mapsto P\varphi(R/P)$ is the perspective of the concave function φ, which is jointly concave on $\mathbb{R} \times \mathbb{R}_{++}$ (Boyd and Vandenberghe, 2004, §3.2.6). So $\ln g$

is jointly concave on D. Since $\ln g \to -\infty$ as $r \to 1$, the supremum over r is attained in $[0,1)$. Partial maximisation of a jointly concave function over a convex set preserves concavity (Boyd and Vandenberghe, 2004, §3.2.5). Hence $\ln v(P) = \max_r \ln g(r,P)$ is concave.

For strict concavity, take $P_1 \neq P_2$ with maximisers r_1, r_2 and $t \in (0,1)$. Under Assumption 4(a), $\partial_r \ln g(0,P) > 0$, so $r_i > 0$. If $r_1 \neq r_2$, strict concavity of $\ln(1-r)$ gives $\ln g(r_t,P_t) > t\ln g(r_1,P_1) + (1-t)\ln g(r_2,P_2)$. If $r_1 = r_2 = r > 0$, then $P \mapsto P\varphi(br/P)$ has second derivative $\varphi''(br/P)(br)^2/P^3 < 0$, which gives the same strict inequality. In either case $\ln v(P_t) \geq \ln g(r_t,P_t) > t\ln v(P_1) + (1-t)\ln v(P_2)$.

(iii) Fix ϱ with $\nu_0(\varrho) > 0$. For small P, $r = \varrho P/b$ is feasible and

$$\frac{v(P)-1}{P} \geq \frac{g(\varrho P/b, P)-1}{P} = \frac{(1-\varrho P/b)\,Q(\varrho P,P)\,e^{-\theta P[\lambda - \Delta\varphi(\varrho)] - \theta_c\lambda_c P} - 1}{P} \longrightarrow \nu_0(\varrho) > 0.$$

So $v > 1 = v(0)$ on some interval $(0,\epsilon)$. A strictly log-concave function is strictly quasi-concave, so its maximiser is unique. It is positive because v exceeds $v(0)$ near zero, and finite because $v \to 0$. Strict quasi-concavity gives strict monotonicity on either side of $\hat{P}$. Since v falls continuously and strictly from $\hat{v}$ to 0 on $[\hat{P},\infty)$, each $F \in (0,\hat{v})$ has exactly one root there.

(iv) For $P > 0$ the maximiser $r^*(P)$ is unique (strict concavity in r) and $\partial_P g$ is continuous. The envelope theorem (Milgrom and Segal, 2002) gives $v'(P) = \partial_P g(r^*(P),P)$. Differentiating $\ln C$,

$$\partial_P\big[-\theta\lambda P + \theta\Delta P\varphi(R/P) - \theta_c\lambda_c P\big] = -\theta\lambda + \theta\Delta\big[\varphi(c) - c\varphi'(c)\big] - \theta_c\lambda_c, \qquad c = R/P,$$

which is (12). Concavity and $\varphi(0) = 0$ give $\varphi(c) - c\varphi'(c) \geq 0$.

(v) $\partial_r\partial_P \ln g = b\,[\partial_R\partial_P \ln Q + \partial_R\partial_P \ln C]$. The first term is nonnegative by Assumption 1. For the second, $\partial_R \ln C = \theta\Delta\varphi'(R/P)$, so $\partial_P\partial_R \ln C = -\theta\Delta\varphi''(c)\,c/P \geq 0$. So $\ln g$ has increasing differences in (r,P), and the unique maximiser is nondecreasing by Topkis's monotonicity theorem (Topkis, 1998). □

Proof of Lemma 5. b enters g only through $R = br$, so by the envelope theorem $\partial \ln v/\partial \ln b = b\,r\,\partial_R \ln(QC)$ at r^*. At an interior optimum the first-order condition is $b\,\partial_R \ln(QC) = 1/(1-r)$, so $\partial \ln v/\partial \ln b = r^*/(1-r^*)$; at a corner $r^* = 0$ both sides are zero. A researcher's payoff is $a\,v(P)$, so production speed raises it by $\mathrm{d}\ln a$. A speed-up by $\mathrm{d}\ln(\text{speed})$ of reading saves $r^*\mathrm{d}\ln b$ of her time and of production saves $x^*\mathrm{d}\ln a$, which gives the value per unit of time saved, $1/x^*$, in both cases. □

Proof of Proposition 1. By Theorem 1(iv), the effect of a marginal paper on an incumbent's per-period payoff is $a\,v'(P)$. Its sign is that of $\hat{P} - P$ by Theorem 1(iii). □

Proof of Proposition 2. With $b_c = \infty$, $n(P) = P/(ax^*(P))$ is continuous and strictly increasing, since x^* is nonincreasing by Theorem 1(v). Let $M(P)$ solve $M\,(n(P) + \xi/s) = N$. It is unique, continuous and strictly decreasing in P. Define $\Psi(P) = \ln v(P) - \ln F(M(P),P)$ on $[\hat{P},\infty)$. Since

$\ln F = -\eta_F \ln(M/\bar{M}) + \ln[(1-\omega)e + \omega v(P)]$,

$$\frac{\mathrm{d}\Psi}{\mathrm{d}\ln P} = (1-\theta_F)\,\varepsilon_v - \eta_F \frac{n\,D}{n+\xi/s} < 0 \quad \text{for } P > \hat{P},$$

because $\varepsilon_v < 0$ there and $\theta_F < 1$. $\Psi(\hat{P}) > 0$ by assumption. As $P \to \infty$, $v \to 0$ while $F \geq (1-\omega)e\,(M/\bar{M})^{-\eta_F}$ is bounded away from zero when $e > 0$, so $\Psi < 0$ eventually. Hence there is a unique root $P^* > \hat{P}$. Crossings on $(0, \hat{P})$ violate stability, since there $v' > 0$. Output per researcher is aF by Lemma 4, and $F = v(P^*) < \hat{v}$. With $\eta_F = 0$, $v(P^*) = (1-\omega)e + \omega v(P^*)$, so $v(P^*) = e$. □

Proof of Proposition 3. In the large-topic limit $M = N/n(P)$, with members per topic $n(P)$ from Lemma 2, which are proportional to $1/a$. So the equilibrium condition $w(P) = F(N/n(P), P)$ involves N and a only through Na, and $\mathrm{d}\ln M = \mathrm{d}\ln(Na) - D\,\mathrm{d}\ln P$, where

$$D = \frac{\mathrm{d}\ln[n\,a]}{\mathrm{d}\ln P} = 1 - \varepsilon_x + \frac{\rho}{\delta}\frac{P/b_c}{1-P/b_c}.$$

Log-differentiating,

$$\varepsilon_w\,\mathrm{d}\ln P = \theta_F \varepsilon_U\,\mathrm{d}\ln P - \eta_F\left[\mathrm{d}\ln(Na) - D\,\mathrm{d}\ln P\right],$$

which gives (13) since $\varepsilon_w < 0$. Then $\mathrm{d}\ln M = \mathrm{d}\ln N - D\,\mathrm{d}\ln P$, and papers per researcher $a\,x^*(P)$ move with elasticity $\varepsilon_x\,\mathcal{E}$. If $\eta_F = 0$, P^* solves $w(P) = (1-\omega)e + \omega\bar{U}_F(P)$, which involves neither N nor a. The overlap rate and the aware share are functions of P^* by Lemma 1. Free entry into science changes P^* only through N. Outside the large-topic limit, $N = M\,(n + \xi/s)$ and D is multiplied by the share of researchers who are members, $n/(n+\xi/s)$. □

Proof of Proposition 4. (i) In the large-topic limit every spell a researcher takes is a spell in a topic of the common density P, worth $a\,w(P)$ by Lemma 2, so output per researcher is $a\,w(P)$ by the renewal-reward argument of Lemma 4. Its maximiser is $\hat{P}_w$, unique by strict log-concavity of w (Proposition 5(i); with $b_c = \infty$, $w = v$ and $\hat{P}_w = \hat{P}$). Since $\hat{P}_w < P^*$, the planner has fewer members per topic and, with N fixed, more topics. (ii) With $\omega = 0$ and $\eta_F = 0$, $E = kM$ with $k = \xi/s$, and output is $a\,M\,(e\,k + n\,w)$ with $M = N/(n+k)$. Maximising $(ek + nw(P(n)))/(n+k)$ over n, the first-order condition is $(w + n\,w_n)(n+k) = ek + nw$, or $k(w-e) = -n\,w_n\,(n+k)$. Since $n\,w_n/w = \varepsilon_w/D$, this is (16). At P^*, $w = F = e$, so the left side of (16) is zero while the right side is positive: the planner lowers density. At $\hat{P}_w$ the right side is zero while the left side is positive when $\hat{w} > e$: she raises it. So $P^o \in (\hat{P}_w, P^*)$. (iii) With a subsidy s^o to explorers the equilibrium condition becomes $w(P) = F(M, P) + s^o$, which $s^o = \hat{w} - F(M^o, \hat{P}_w)$ makes hold at $\hat{P}_w$. (iv) Let the planner also set x. Per-member output is $a\,g(x, P)\,(1 - P/b_c)^{\rho/\delta}$ and, at fixed membership, $\partial \ln P/\partial \ln x = 1/D_m$ with $D_m = 1 + (\rho/\delta)(P/b_c)/(1 - P/b_c)$. The first-order condition for x is

$$\partial_x \ln g + \left[\partial_P \ln g + (\ln\Gamma)'(P)\right]\frac{P}{x\,D_m} = 0,$$

where we use the newcomer's survival factor for the value of membership. At the private optimum $\partial_x \ln g = 0$ and, by the envelope theorem, $\partial_P \ln g + (\ln \Gamma)'(P) = (\ln w)'(P)$. The wedge therefore has the sign of $(\ln w)'$, and it vanishes at $\hat{P}_w$. □

Proof of Proposition 5. (i) $\ln \Gamma = \nu \ln(1 - P/b_c)$ has second derivative $-\nu/(b_c - P)^2 < 0$, so $\ln w = \ln v + \ln \Gamma$ is strictly concave on $(0, b_c)$, and it tends to $-\infty$ at b_c. Since $(\ln w)'(\hat{P}) = (\ln \Gamma)'(\hat{P}) < 0$, the peak of w lies below $\hat{P}$ when $\hat{P} < b_c$. (ii) The equilibrium condition and stability are those of Definition 1; $w(b_c) = 0 < F$ puts the falling-side root below b_c. The premium follows from $v(P^m)\Gamma(P^m) = F$ and $-\ln \Gamma = (\rho + \xi)T$. (iii) If $\hat{P} < b_c$, $P^m < \hat{P}$ iff $w(\hat{P}) < F$, since w is falling on $(\hat{P}_w, b_c)$; this reads $\ln \hat{v} + \nu \ln(1 - \hat{P}/b_c) < \ln F$. If $\hat{P} \geq b_c$, $P^m < b_c \leq \hat{P}$. (iv) From $\ln \zeta + \ln v(P^m) + \ln \Gamma(P^m) = \ln F$, differentiate with respect to $\ln P^m$. With $x = P^m/b_c$, the function $x/(1-x) + \ln(1-x)$ is zero at $x = 0$ and has derivative $x/(1-x)^2 > 0$, so $\varepsilon_\Gamma > -\ln \Gamma$. □

Proof of Theorem 2. (i) By Lemma 2, $T = \infty$ for any positive stock when $P \geq b_c$. The value of joining is then zero whatever $v(P)$ is, so no newcomer joins and incumbents' payoffs are not competed away. (ii) w is continuous and positive on $[0, b_c)$ with $w(0) = 1$ and $w \to 0$ as $P \to b_c$. From $\zeta w(P^m(\zeta)) = F$, $w(P^m) = F/\zeta \to 0$ as $\zeta \to \infty$, which forces $P^m(\zeta) \to b_c$. With $x = P^m/b_c \to 1$, the premium $-\nu \ln(1 - x)$ and the gradient $\nu x/(1 - x)$ diverge, and their ratio $x/[-(1 - x)\ln(1 - x)] \to \infty$. (iii) Along the growth path the stock is positive for $P > P_0$, so the myopic value $v(P)[1 + \delta S/(b_c - P)]^{-\nu}$ tends to zero as $P \to b_c$; entry stops at a density where it equals $F > 0$, below b_c. (iv) With fixed membership and reading shares, (1) scales every flow by $1 + \alpha$. A topic whose new flow is at least b_c is closed by (i). With no entry, membership falls at rate ρ, and the flow falls at the same rate at fixed reading shares, so it stays at or above b_c for $\ln[(1 + \alpha)P^m/b_c]/\rho$ periods. □

Proof of Corollary 1. Total member output in a topic is $n\, a\, w(P(n))$, so one more member changes the output of the others by $n\, a\, w'(P)\, \mathrm{d}P/\mathrm{d}n$. Here $\mathrm{d}P/\mathrm{d}n > 0$ and $w' < 0$ at a stable equilibrium. Writing $w' = v'\Gamma + v\Gamma'$ separates the spillover on incumbents from the catch-up burden. The ordering $\hat{P}_w < P^m < \hat{P}$ combines Proposition 5(iii) with stability. The loss follows from Proposition 4: per-researcher output is aF in equilibrium and $a\hat{w}$ at the optimum. □

Proof of Proposition 6. (i) At private choices, $\ln s(P) = \ln g(r^*(P), P) - \theta P[(\lambda_s - \lambda) - (\Delta_s - \Delta)\varphi(br^*/P)]$. Differentiate in P. The derivative of $\ln g(r^*(P), P)$ is v'/v by the envelope theorem. The direct derivative of the bracket term is $-\theta[(\lambda_s - \lambda) - (\Delta_s - \Delta)(\varphi - c\varphi')]$. Because researchers do not maximise $\ln s$, the bracket term also changes through r^*, by $\theta b(\Delta_s - \Delta)\varphi'(c)\, r^{*\prime}(P)$. The effect of one more member on others' social output is $n\, a\, s'_w(P)\, \mathrm{d}P/\mathrm{d}n$, as in Corollary 1. (ii) With $\Delta_s = \Delta$ the bracket term is $-\theta(\lambda_s - \lambda)P$, independent of r, so $s = ve^{-\theta(\lambda_s - \lambda)P}$ and $s_w = we^{-\theta(\lambda_s - \lambda)P}$. In the large-topic limit the social planner maximises s_w, which is strictly log-concave. Its peak lies below P^m if and only if $(\ln s_w)'(P^m) < 0$, that is $(\ln w)'(P^m) < \theta(\lambda_s - \lambda) = -\tau\theta\lambda$. Since

$(\ln w)'(P^m) = \varepsilon_w(P^m)/P^m < 0$, this is $\tau < |\varepsilon_w(P^m)|/(\theta\lambda P^m)$. (iii) With $\ln C_s = (1-\tau)\ln C$, $\ln s = \ln v - \tau \ln C^*$, and the planner maximises $\ln s + \ln \Gamma$. □

Proof of Theorem 3. Index the tool by τ, so that $w = w(P;\tau)$, $F = F(M,P;\tau)$ and $n = n(P;\tau)$. The equilibrium condition is $\ln w = \ln F$, and in the large-topic limit $M = N/n$. Let $h = \partial_\tau \ln w$ and $m = \partial_\tau \ln n$ at fixed P. Let f be the derivative of $\ln F$ in τ at fixed P when M follows $N/n(P;\tau)$; it is the direct effect plus $\eta_F m$. Differentiating,

$$\varepsilon_w \,\mathrm{d}\ln P + h = f + \theta_F \varepsilon_U \,\mathrm{d}\ln P + \eta_F D \,\mathrm{d}\ln P,$$

so $\mathrm{d}\ln P = (h-f)/\Psi$. By Lemma 4, $Y/N = aF$, and in equilibrium the total change in $\ln F$ is $f + (\theta_F\varepsilon_U + \eta_F D)\,\mathrm{d}\ln P = f + (\Psi - |\varepsilon_w|)(h-f)/\Psi$, which gives (17). For (18), $\mathcal{E} = \eta_F/\Psi$ by Proposition 3, so $\Psi(1-\mathcal{E}D) = |\varepsilon_w| + \theta_F\varepsilon_v$ and $\lambda_F = (1-\mathcal{E}D)|\varepsilon_w|/(|\varepsilon_w| + \theta_F\varepsilon_v)$. When mature topics lie on the rising side of v ($\varepsilon_v \geq 0$), the fraction is at most one and $1 - \lambda_F \geq \mathcal{E}D$. □

Proof of Corollary 2. (i) a does not enter v or Γ, so $h = 0$, and $n \propto 1/a$ at fixed P, so $m = -\mathrm{d}\ln a$ and $f = -\eta_F \,\mathrm{d}\ln a$. Then $\mathrm{d}\ln(Y/N) = \mathrm{d}\ln a\,(1-\lambda_F\eta_F)$, and $\lambda_F\eta_F = \mathcal{E}|\varepsilon_w|$. (ii) A learning tool raises $\ln v$ by h_v at every P, hence $\ln w$ by h_v and $\ln \bar{U}_F = \ln v(P)$ by h_v, so the direct effect on $\ln F$ is $\theta_F h_v$; $m = -\partial_\tau \ln x^*$ is second order for the tools we consider. Then $\Pi_L = \lambda_F\theta_F + 1 - \lambda_F$. (iii) A catch-up tool raises $\ln\Gamma$ only; founders' payoffs do not involve Γ, so the direct effect on F is zero and $\Pi_C = 1 - \lambda_F$. (iv) $\Pi_L - \Pi_C = \lambda_F\theta_F \geq 0$ and $1 - \mathcal{E}|\varepsilon_w| - \Pi_L = \lambda_F(1-\theta_F-\eta_F)$. The per-hour statement combines these with Lemma 5: reading saves $r^*\mathrm{d}\ln b$ of time and is worth $\Pi_L\, r^*\mathrm{d}\ln b/x^*$ in the long run; production saves $x^*\mathrm{d}\ln a$ and is worth $(1-\mathcal{E}|\varepsilon_w|)\,\mathrm{d}\ln a$. □

Proof of Proposition 7. (i) and (ii) restate Proposition 3 and Corollary 2(i): P^* depends on Na; output per researcher is aF; papers per researcher are $a\,x^*(P^*)$. Under free entry into science $aF = u_0 + c$. For (iii)–(iv), $\partial_b \ln g = r[\partial_R \ln Q + \partial_R \ln C] \geq 0$. Q is increasing in β. C is increasing in σ and decreasing in ℓ, and ℓ is increasing in κ_p. Γ is increasing in b_c. So w rises pointwise by h while F rises by at most $\theta_F h < h$, and Theorem 3 gives $\mathrm{d}\ln P^* > 0$; $M^* = N/n(P^*)$ falls because n is increasing. The speed limit is b_c itself. The overshooting threshold falls with b_c because g^*_∞ in (20) is decreasing in b_c and the myopic value on the growth path is increasing in b_c at every (P,S). (v) A in (4) is increasing in φ, and φ is increasing in σ. □

Proof of Proposition 8. In a topic, n_p producing members read $n_p\, b\, r^*$ papers per period and write $P = n_p\, a\, x^*$, so reads per paper are $\mathcal{A} = (b/a)\, r^*/(1-r^*)$. Then $\ln Y_\alpha = \ln Y + \alpha \ln \mathcal{A}$. For a production tool, $\mathrm{d}\ln(Y/N) = (1-\mathcal{E}|\varepsilon_w|)\,\mathrm{d}\ln a$ by Corollary 2 and $\mathrm{d}\ln\mathcal{A} = -\mathrm{d}\ln a + \varepsilon_r \mathcal{E}\,\mathrm{d}\ln a$ with $\varepsilon_r = \mathrm{d}\ln[r^*/(1-r^*)]/\mathrm{d}\ln P$, so with $\alpha = 1$ the total is $\mathcal{E}(\varepsilon_r - |\varepsilon_w|)\,\mathrm{d}\ln a$, a product of small numbers. For a reading tool, $\mathrm{d}\ln\mathcal{A} = \mathrm{d}\ln b$ plus the change in $\ln[r^*/(1-r^*)]$, and $\mathrm{d}\ln(Y/N) = \Pi_L\, r^*\mathrm{d}\ln b/x^*$ is small when reading takes little time. □

Proof of Remark 4. Let the cross-section of topics have researcher weights $\omega(a)$ over ages a, so that the measured log density is $L = \int \omega(a) \ln P(a)\, \mathrm{d}a$. If the area's growth rate rises by $\mathrm{d}g$ for T years, the cohorts born in the window are larger by the factor $e^{\mathrm{d}g(T-a)}$ relative to older ones, so ω becomes $\omega(a)e^{\mathrm{d}g(T-a)^+} / \int \omega e^{\mathrm{d}g(T-a)^+}$. To first order, $\mathrm{d}L = \mathrm{d}g\, \mathrm{Cov}_\omega(\ln P, (T-a)^+)$. Log volume rises by $T\mathrm{d}g$ more than it would have, so the composition term in the measured elasticity is $\mathrm{Cov}_\omega(\ln P, (T-a)^+)/T$. Mature densities depend on N only through the frontier and adjust at once, which gives (15). A constant growth rate gives a constant age composition, which area fixed effects absorb. □

Proof of Proposition 9. On the growth path,

$$S(t) = \int_0^t P_0 e^{gu} e^{-\delta(t-u)}\, \mathrm{d}u = P(t)\, \frac{1 - e^{-(g+\delta)t}}{g+\delta},$$

and $e^{-(g+\delta)t} = (P_0/P)^{(g+\delta)/g}$, which gives (19). At a given P, S is decreasing in g: both $1/(g+\delta)$ and $1-(P_0/P)^{1+\delta/g}$ fall as g rises.

(i) The myopic value $V(P;g) = v(P)[1 + \delta S(P;g)/(b_c - P)]^{-\nu}$ is increasing in g at every $P \in (P_0, b_c)$, so the first P at which it falls to F is nondecreasing in g; it is below b_c by Theorem 2(iii). Entry starts because $V(P_0;g) = v(P_0) > F$. As $g \to 0$, $S(P;g) \to P/\delta$ uniformly on compact subsets of (P_0, b_c), so $V \to w$; if $w(P_0) < F$, $w < F$ on (P_0, P_ℓ), where $P_\ell < P^m$ is the rising-side root of $w = F$, and slow topics stop in that interval. As $g \to \infty$, $S \to 0$ and $V \to v$, which exceeds F on $[P_0, \bar{P}_F)$ with $\bar{P}_F > P^m$, so fast topics pass P^m. By monotonicity there is a threshold g_0 between the two. If entry stops at $P_s < P^m$, then $P_s < P_\ell$: on $[P_\ell, P^m]$, $w \geq F$ and $V > w$ because the stock along the path is below the stationary stock. (After such a stop the flow falls at rate $\rho < \delta$ and the ratio of stock to flow rises towards $1/(\delta - \rho) > 1/\delta$; we verify numerically that, for the calibrated parameters, the value of joining never returns to F.)

(ii) The topic overshoots iff $V(\cdot; g) > F$ on $[P_0, \hat{P}]$. The set of such g is an upper set by monotonicity, so overshooting is characterised by a threshold g^*. It requires taking off ($g \geq g_0$) and $V(\hat{P}; g) > F$ ($g > g_{\hat{P}}$). Conversely, if $g > \max\{g_0, g_{\hat{P}}\}$, $V > F$ on $[P_0, P^m]$ and at $\hat{P}$; on $[P^m, \hat{P}]$, $\ln V$ is concave in P when the stock is $P/(g+\delta)$, since $\ln v$ is concave and $-\ln[1 + \delta P/((g+\delta)(b_c - P))] = -\ln[b_c - (1-c)P] + \ln(b_c - P)$ with $c = \delta/(g+\delta) < 1$ is concave, so $V > F$ on the whole interval; with the exact stock we verify this numerically for the calibrated parameters. Hence $g^* = \max\{g_0, g_{\hat{P}}\}$. Because the exact stock is below $P/(g+\delta)$, $V(\hat{P}; g)$ exceeds its value at the bound, so $g_{\hat{P}}$ is at most the rate that solves $\hat{v}[1 + \delta\hat{P}/((g+\delta)(b_c - \hat{P}))]^{-\nu} = F$, which is (20).

(iii) Suppose entry stops at $P_s \in [P^m, \hat{P}]$. Afterwards the flow falls from P_s to P^m, so a newcomer at t_s faces flows in $[P^m, \hat{P}]$, where $v \geq v(P^m)$, and reads less than she would at the constant flow P_s: her catch-up time is at most $T(P_s, S_s) \leq T(\hat{P}, \hat{P}/(g+\delta))$, since T is increasing in both arguments and $S_s \leq S(\hat{P}) \leq \hat{P}/(g+\delta)$. Her value is at least $v(P^m)\,[1 + \delta\hat{P}/((g+\delta)(b_c - \hat{P}))]^{-\nu}$. A newcomer to the mature topic earns $F = v(P^m)(1 - P^m/b_c)^\nu = v(P^m)[1 + P^m/(b_c - P^m)]^{-\nu}$. If $g > g^*_{RE}$, then $\delta\hat{P}/((g+\delta)(b_c - \hat{P})) < P^m/(b_c - P^m)$, so the newcomer at t_s earns more than F,

contradicting that entry stops. Hence $P_s > \hat{P}$.

(iv) Incumbents have already paid the catch-up cost. At the stopping date a newcomer's value equals F after a survival factor below one, so an incumbent's continuation value exceeds F, and incumbents stay. With no entry, membership decays at rate ρ. Using the stationary stock, entry resumes when $w(P) = F$, at P^m. □

Proof of Proposition 10. By Lemma 4, $Y = NaF$, and in equilibrium $\ln F = \ln w(P^*)$, so $\mathrm{d}\ln Y/\mathrm{d}\ln N = 1 + \varepsilon_w \mathcal{E}$. Holding M fixed, $N = M\, n(P)$ gives $\mathrm{d}\ln P = \mathrm{d}\ln N/D$, and output per researcher is $a\, w(P)$ by the renewal-reward argument, so $\partial \ln Y/\partial \ln N|_M = 1 + \varepsilon_w/D$. The bound with fertility depletion is proved in Appendix B.1. □

Proof of Proposition 11. (i) The first follower of a newborn topic reads nothing and earns $v(P_0)$; she joins only if this exceeds the value of exploring, $F + s$, or, with a flat entry tax, if $v(P_0) - s > F$. (ii) The tax τS is zero at birth, so (i) does not apply. A newcomer to a stationary topic pays $\tau P/\delta$, so mature topics solve $w(P) - \tau P/\delta = F$ on the falling side of the left-hand side, whose peak lies below $\hat{P}_w$ because the tax rises with P. At τ^o the equation holds at $\hat{P}_w$. □

B Extensions

B.1 Fertility depletion

Suppose topics also differ in fertility, and later topics are less fertile. The m-th topic opened has fertility $f(m) = \bar{f}\, m^{-\eta_f}$, so members of topic m earn $af(m)$ times the payoff curve. The value of exploring falls with the number of open topics at elasticity η_a, which captures *access depletion*: the η_F of the main text. A successful explorer opens a topic of fertility $f(M)$. Newcomers are indifferent between topic m and exploring when

$$f(m)\, w(P_m) = F(M) f(M) \iff w(P_t) = \bar{F}\, M^{-\eta_a}\, t^{\eta_f}, \qquad t \equiv m/M \in (0, 1],$$

where for simplicity founders' payoffs are absorbed into $\bar{F}$ ($\theta_F = 0$).

Proposition 12 (Scale invariance under fertility depletion)**.** *(i) If $\eta_a = 0$, the distribution of densities across topics, and hence of overlap rates and aware shares, is invariant to N and a, whatever the value of η_f. (ii) In general, with $\mathcal{D} \equiv \mathrm{d}\ln M^*/\mathrm{d}\ln N$,*

$$\lambda_J = 1 - (\eta_a + \eta_f)\, \mathcal{D}, \qquad \mathcal{D} = \Big(1 + \eta_a\, \mathbb{E}_n\Big[\frac{D(P_t)}{|\varepsilon_w(P_t)|}\Big]\Big)^{-1},$$

where $\mathbb{E}_n$ averages over topics with weights proportional to members. The paper-weighted volume elasticity of density identifies η_a, not η_f, and $\lambda_J \le 1 - \eta_a \mathcal{D}$, the value obtained when $\eta_f = 0$. With a single density, $1 - \eta_a \mathcal{D} = 1 - \mathcal{E}|\varepsilon_w|$.

Proof. (i) With $\eta_a = 0$ the indifference condition pins P_t as a function of the relative rank t alone. Members per topic $n(P_t)$ are then invariant, so $M = N/\int_0^1 n(P_t)\mathrm{d}t$ scales with N. (ii) Every researcher's career payoff is $aF(M)f(M)$, so $Y = Na\bar{F}\bar{f}M^{-\eta_a-\eta_f}$ and $\lambda_J = 1-(\eta_a+\eta_f)\mathcal{D}$. At a fixed rank, $\varepsilon_w(P_t)\,\mathrm{d}\ln P_t = -\eta_a\,\mathrm{d}\ln M$. Substituting into $N = M\int_0^1 n(P_t)\mathrm{d}t$ gives $\mathrm{d}\ln N = \mathrm{d}\ln M\,(1+\eta_a\mathbb{E}_n[D/|\varepsilon_w|])$. □

Fertility depletion, the textbook "fishing out" of ideas (Kortum, 1997), thus leaves the anchor result intact and strengthens it: volume opens new topics exactly. It adds to stepping on toes a component that density data cannot see. This is why Proposition 10 delivers an upper bound on λ_J.

B.2 Fertility heterogeneity and the cross-section

With fertility ζ drawn independently at a topic's birth, a mature topic solves $\zeta w(P^m) = F$. More fertile topics are denser, their incumbents earn more, and the cross-sectional returns gradient is $\varepsilon_\Gamma(P^m)$ (Proposition 5(iv)). The most fertile topics sit closest to the speed limit (Theorem 2). Topics with $\zeta\hat{w} < F$ cannot sustain replacement entry. They shrink once their growth spurt ends and are eventually abandoned, like topics that grow too slowly to take off (Proposition 9).

B.3 Nested topics and the radius pattern

Fact 1 finds that the volume elasticity of density rises with the similarity radius: it is close to zero at the radius of close overlap and larger at broad radii. The model's topic is a neighbourhood of fixed size (Remark 1), and a two-scale version delivers the pattern. Let idea space consist of *fields* of fixed breadth, each containing topic cells. A researcher can join an open topic, open a new topic inside an existing field, or open a new field. Opening a topic inside a field is worth $F_T(m) = \bar{F}_T\,m^{-\eta_T}$, where m is the number of open topics per field, and opening a field is worth $F_B(M_B) = \bar{F}_B\,M_B^{-\eta_B}$, where M_B is the number of fields. Learning and covering act within topics.

Proposition 13 (The radius pattern)**.** *In the large-topic limit, the volume elasticity of density at the topic scale is*

$$\mathcal{E}_T \equiv \frac{\mathrm{d}\ln P^*}{\mathrm{d}\ln N} = \frac{1}{D+|\varepsilon_w|\,(1/\eta_T+1/\eta_B)},$$

and the volume elasticity of density at the field scale, papers per unit of field breadth mP^*, is

$$\mathcal{E}_B \equiv \frac{\mathrm{d}\ln(m^*P^*)}{\mathrm{d}\ln N} = \mathcal{E}_T\Big(1+\frac{|\varepsilon_w|}{\eta_T}\Big) \geq \mathcal{E}_T.$$

If topics are easy to open inside fields and fields hard to open ($\eta_T \ll \eta_B$), $\mathcal{E}_T$ is close to zero while $\mathcal{E}_B$ is close to $1/(1+\eta_T/\eta_B)$: close overlap is pinned while broad neighbourhoods fill. With neither margin open ($\eta_T, \eta_B \to \infty$) both equal $1/D$, the fixed-field benchmark.

Proof. Indifference among the three options gives $w(P) = F_T(m) = F_B(M_B)$ and the adding-up condition is $N = M_B\,m\,n(P)$. Log-differentiating, $\varepsilon_w\,\mathrm{d}\ln P = -\eta_T\,\mathrm{d}\ln m = -\eta_B\,\mathrm{d}\ln M_B$

and $\mathrm{d}\ln N = \mathrm{d}\ln M_B + \mathrm{d}\ln m + D\,\mathrm{d}\ln P$. Substituting $\mathrm{d}\ln m = (|\varepsilon_w|/\eta_T)\,\mathrm{d}\ln P$ and $\mathrm{d}\ln M_B = (|\varepsilon_w|/\eta_B)\,\mathrm{d}\ln P$ gives $\mathcal{E}_T$, and $\mathrm{d}\ln(mP) = (1 + |\varepsilon_w|/\eta_T)\,\mathrm{d}\ln P$ gives $\mathcal{E}_B$. The limits follow directly. □

The proposition turns the radius pattern of Fact 1 into a prediction and a measurement. The ratio of the broad-radius to the fine-radius elasticity identifies $1 + |\varepsilon_w|/\eta_T$, the ease of opening topics within existing fields. The ambitious version of this extension is a continuous idea space with a wide learning kernel and a narrow covering kernel, in which topics form endogenously as in the spatial models of Lucas and Rossi-Hansberg (2002) and Allen and Arkolakis (2014). A wide learning kernel adds a spillover from neighbouring topics, which makes opening a topic next to existing ones more attractive than opening one in empty space, and so raises $\mathcal{E}_B$ relative to $\mathcal{E}_T$. We leave that version to future work.

B.4 Learning from the stock

In the main text learning draws on the flow and catch-up on the stock. Let learning instead draw on the effective literature $L = (1-\chi)P + \chi\,\delta S$, with $\chi \in [0,1]$, while covering papers still arrive with the flow. In a stationary topic $\delta S = P$, so $L = P$, and every stationary result is unchanged: the payoff curve, the mature density, the speed limit and the incidence of tools. Along a growth path the stock lags, so $L < P$ and a young topic offers less to learn from than a mature topic of the same density. Two things follow. A newborn topic, which has no stock, offers its first follower v evaluated at $L = (1-\chi)P_0$, which is below $v(P_0)$; the condition for newborn topics to attract followers tightens. In the calibration it fails for every χ above 0.11: if more than that share of learning drew on the accumulated stock, no new topic would ever find a follower. Below that bound young topics are less attractive, so they need to grow faster to take off and to overshoot. With $\chi = 0.06$ the take-off threshold rises from 2.2% to 2.6% and the overshooting threshold from 20.5% to 21.2%. Learning from the flow is therefore not a convenience: it is what lets topics be born.

B.5 Search for overlap on every project

In the main text a researcher's reading protects all her projects at once. Suppose instead that overlap must be checked project by project, as our evidence on targeted search suggests. Each project takes $1/a$ of production time plus a search time t in which the researcher screens $b_s t$ papers of her topic's literature, so that each earlier covering paper is caught with probability $\varphi(b_s t/P)$. Learning still comes from the reading share r. Her payoff per period is

$$a\,\frac{(1-r)\,Q(br,P)}{1+a\,t}\,\exp\Big\{-\theta P\big[\lambda - \Delta\varphi(b_s t/P)\big] - \theta_c\lambda_c P\Big\}.$$

Let $\varsigma \equiv at/(1+at)$ be the share of project time spent searching.

Proposition 14 (Per-project search)**.** *(i) At a fixed density, a production speed-up raises the payoff by* $(1-\varsigma)\,\mathrm{d}\ln a$ *and a search speed-up by* $\varsigma\,\mathrm{d}\ln b_s$*: per unit of time saved, every speed-up is still worth the same. (ii) Faster production lowers the papers screened per project,* $b_s t$*, and so raises the share of earlier covering papers that go undetected. (iii) Production speed no longer drops out of the payoff curve:* v *depends on* a *through* ς*.*

Proof. (i) By the envelope theorem, $\partial\ln(\text{payoff})/\partial\ln a = 1 - at/(1+at)$ at the optimal t, and $\partial/\partial\ln b_s = \theta\Delta\varphi'(c_s)c_s$ with $c_s = b_s t/P$, which the first-order condition for t, $\theta\Delta\varphi'(c_s)\,b_s/P = a/(1+at)$, equates to $at/(1+at)$. Time saved per project is $(1/a)\mathrm{d}\ln a$ and $t\,\mathrm{d}\ln b_s$ respectively, and both are valued at the same shadow price. (ii) Write the first-order condition as $G(t,a) \equiv \theta\Delta\varphi'(b_s t/P)\,b_s/P - a/(1+at) = 0$. Then $\partial G/\partial a = -1/(1+at)^2 < 0$, and $\partial G/\partial t < 0$ by the second-order condition, so $\mathrm{d}t/\mathrm{d}a = -(\partial G/\partial a)/(\partial G/\partial t) < 0$: search per project falls, and with it detection. (iii) follows from (i). □

Per-project search narrows the gap between production and reading tools at fixed topics, since production is only a share $1-\varsigma$ of a project, and it makes faster production raise the rate of unaware duplicates. The incidence results of Section 6 are unchanged in form: what matters for the long run is still how a tool moves the frontier relative to existing topics.

B.6 Heterogeneous ability

Ekerdt and Wu (2024) attribute part of measured stepping on toes to falling researcher ability. Let ability z multiply a researcher's credit. If ability and reading are complements (Q increasing in $z\,R$), able researchers gain more from dense literatures and sort into them. If ability also raises the explorer's founding rate, they sort to the frontier, where they also raise θ_F. Which force wins determines whether the dense-topic premium in Fact 2 partly reflects sorting. Author fixed effects remove level differences in ability but not ability-specific returns to density. The two can be separated by interacting density with pre-period ability (early-career citations), which we leave for future work.

B.7 Credit rules

Partial credit enters only through $\Theta = \theta\lambda$, Δ/λ and $\Theta_c = \theta_c\lambda_c$. Theorem 1 requires $\lambda > \ell$ and nothing else. The sign of the externality at a stable equilibrium (Proposition 2 and Corollary 1) therefore survives whenever duplicates earn some credit. Concurrent overlap adds a term linear in density that no reading prevents: it steepens the fall of v at high density without changing what reading is worth for protection. Replication value can be added as a positive social value of aware follow-ons that is not credited privately. This raises the social value of detection relative to the private one, but it does not change the sign of w' at equilibrium.

B.8 Transitional dynamics after a production shock

A one-time increase in a raises every topic's flow immediately, before membership adjusts. Topics jump to the right along v, overlap rises, and incumbents' payoffs change by $\varepsilon_v \, \mathrm{d} \ln a$. The newcomer's payoff falls, by $|\varepsilon_w| \, \mathrm{d} \ln a$ at the old membership, so newcomers go to the frontier and topics thin through turnover, returning towards P^* at rate ρ. Topics that the shock pushes past the speed limit close entirely until their flow falls back below b_c (Theorem 2(iv)). In the calibration, closing even the densest mature topics requires production speed to rise by a factor of 3.0; a fourfold rise closes topics that hold 32% of mature researcher-years. The overlap rate spikes and decays with half-life $\ln 2/\rho$. Papers per author rise immediately and stay higher. The author-level estimate of Fact 5, about 7% more papers for adopters, bounds the size of the production shock so far. With adopters a minority of active submitters (about one in fifteen in our classification), the aggregate increase in effective production speed is at most a few percent, too small to produce a visible spike in overlap or to close any topic.

C Numerical methods and verification

Solving the researcher's problem. For each $P > 0$ the first-order condition $\partial_r \ln g(r, P) = 0$ has a unique root in $[0, 1)$ because $\ln g$ is strictly concave in r. We find it by vectorised bisection (60 steps) followed by three Newton steps, and set $r^* = 0$ when $\partial_r \ln g(0, P) \leq 0$. The derivative v'/v is computed from the envelope formula (12), not by differencing. Peaks and crossings use bounded Brent searches on the log-concave curves, below the speed limit b_c when it is finite.

Catch-up. Catch-up times use the closed form (7). For a flow that thins after entry stops, the backlog (6) has a closed form on each phase (thinning at rate ρ, then constant at P^m), and the catch-up time is found by bracketing on the thinning phase and solving exactly on the constant one. The closed forms are checked against direct integration of (6).

Stationary equilibrium, incidence and planner. The equilibrium is the root of $\ln w(P) - \ln F(M(P), P)$ on $[\hat{P}_w, b_c)$, which is strictly decreasing there (proof of Proposition 2). Long-run effects of tools re-solve the equilibrium exactly. In each counterfactual explorers' output e is reset so that the calibrated mature density is an equilibrium with $M = 1$, and η_F so that the volume elasticity is 0.15, for each founders' probability ω. The finite-topic planner maximises output over P along the resource constraint with births equal to deaths.

Life cycles. The growth phase has closed forms, $P = P_0 e^{gt}$ and (19). The stopping time is found on a fine time grid and refined by linear interpolation. Topics that stop above their mature density consolidate at rate ρ and then stay mature; topics that stop below it, or whose first follower would not beat the entry bar, thin at rate ρ and are abandoned. For the calibrated parameters we check that a stalled topic is never worth joining again by integrating its stock after the stop

(largest entry gap -0.001 in logs). Diffusion rates and fertility are lognormal, truncated at ±2.5 log standard deviations and integrated on evenly spaced nodes. Because mature density is very elastic with respect to fertility ($1/|\varepsilon_w| \approx 15$), untruncated tails, where a few extremely fertile topics carry many researchers, make the moments converge very slowly. Topic-years are weighted by producing members, $P/x^*(P)$, and by topic survival, $e^{-\xi t}$. Observed log density adds classical measurement error with a standard deviation of 0.3, integrated by seven-point Gauss–Hermite quadrature. Besides matching the noisy arXiv proxy, this makes the quintile moments smooth in the parameters. The search uses a coarse grid (15 diffusion rates, 15 fertility levels, $\Delta t = 0.1$). Candidates are re-ranked, and all reported moments computed, on a fine grid (41 diffusion rates, 51 fertility levels, $\Delta t = 0.05$).

Founders and life-cycle welfare. Founders' payoffs $\bar{U}_F$ are the type-weighted discounted averages of $\zeta v(P(t))$ along simulated life cycles. The founders' fixed point $F = (1-\omega)e + \omega\bar{U}_F(F)$ is solved by Brent's method, re-simulating the life cycle at each trial F. Output per researcher-year on the life cycle counts, per topic born, $1/s$ explorer-years producing e, producing members $P/(ax^*)$ earning $\zeta v(P)$, and members in catch-up, imputed from the flow of new producers $G = \dot{n}_p + \rho n_p$ as $G\,(e^{\rho T} - 1)/\rho$, where T is the catch-up time they face; in a mature topic this reproduces the producing share $(1 - P/b_c)^{\rho/\delta}$ exactly.

Verification. The file `code/tests/test_propositions.py` checks every analytical result numerically:

- log-concavity, single-peakedness, $v(0) = 1$ and $v \to 0$ on 25 random parameter draws satisfying Assumptions 1 to 4, with and without concurrent overlap;
- the envelope formula against finite differences, the monotonicity of $r^*(P)$ (Theorem 1), and the initial slope $v'(0^+) = \sup_\varrho \nu_0(\varrho)$ (Assumption 4(b));
- the time-share value of reading, $\partial \ln v/\partial \ln b = r^*/(1-r^*)$, and the equal value of every speed-up per unit of time saved (Lemma 5);
- the externality decomposition and its sign (Proposition 1);
- the catch-up time (7) against direct integration of the backlog, the speed limit, and the producing share of members (Lemma 2);
- the value of the frontier (10) by Monte Carlo of exploring and founding spells (Lemma 3);
- existence, uniqueness, the falling-side location and dissipation of equilibrium with and without founders and catch-up (Proposition 2);
- the volume elasticity (13) with founders against finite differences, exact invariance to N and a when $\eta_F = 0$, and both stepping-on-toes elasticities (Propositions 3 and 10);
- the planner's ordering $\hat{P}_w < P^o < P^*$ (Proposition 4);

- the cross-topic returns gradient $\varepsilon_\Gamma(P^m)$ and its excess over the premium (Proposition 5), and the speed limit: mature density below b_c, unbounded premia as fertility rises, myopic stopping below b_c, and closure after production shocks (Theorem 2);
- overcrowding with rising returns (Corollary 1);
- the monotonicity of the stopping density, the take-off threshold and the valley in which slow topics stall, the overshooting threshold $g^* = \max\{g_0, g_{\hat{P}}\}$, the closed-form bound (20), and that stalled topics are never worth joining again (Proposition 9(i)–(ii));
- under forward-looking entry, the bound g^*_{RE}, the value at the mature density, and deeper overshooting than under myopia for fast topics (Proposition 9(iii));
- the frontier-incidence formula (17) against exact re-solved equilibria for learning, catch-up and production tools, with and without founders and frontier depletion; the ordering of Corollary 2; and the lower bound (18) (Theorem 3);
- reads per paper and their fall with production speed (Proposition 8);
- the flat subsidy that stops newborn topics from taking off, and the stock-proportional tax that targets the peak of w without affecting newborn topics (Proposition 11);
- the imputation of catch-up members in the life-cycle accounts;
- the social-value identities (Proposition 6) and the composition bias of the short-run volume elasticity (Remark 4);
- a Monte Carlo check of the credit formula (3) with earlier and concurrent overlap (Lemma 1).

All checks pass.

D Data and empirical designs

All scripts are in `empirics/` and run in numerical order. Raw data are public.

The arXiv panel. The source is the arXiv metadata snapshot distributed by Cornell University on Kaggle, file dated 18 October 2026 (Cornell University, 2026). We keep the 2,214,380 papers with new-style identifiers from January 2015 (`01_parse_snapshot.py`). For each paper we record the submitter's name, the parsed author list, the categories (the first is the primary category), title, abstract, and the timestamp of version 1, which is the submission time in UTC. The last submission in the snapshot was made at 13:59:59 ET on 15 October 2026, just before that day's cutoff.

Announcement batches and listing position (`02_listing_position.py`). Each paper is assigned the deadline it was submitted against: the same weekday's cutoff if submitted before it, otherwise the next weekday's. Weekend submissions belong to Monday's batch. The cutoff was

16:00 ET until 30 December 2016 and 14:00 ET from 2 January 2017. Both are visible as spikes in the distribution of submission times. Identifiers are assigned at announcement, so the smallest identifier among the papers with a given deadline marks the start of that announcement block. Every paper is placed in the block its identifier falls in. Papers placed in a later block than their deadline were held for moderation (6.1%). Within blocks, the rank correlation between identifier and submission time is 0.98 for papers not held (median over 3,065 blocks). Position is the rank by identifier within block and primary category. Block starts are monotone except at seven holiday periods, which we drop: 17 December to 8 January and the week of US Thanksgiving.

Cutoff design (`03_rd_sample.py`, `07_rd.py`). Weekday submissions within 180 minutes of the cutoff, deadlines from 9 January 2017 to 16 December 2022 (272,707 papers). The running variable is minutes from the cutoff. Estimates are local linear with separate slopes, a triangular kernel, category-year and weekday fixed effects, and standard errors clustered by submission day. The density of the running variable is discontinuous by construction, since authors bunch in the first seconds after the cutoff, so the main specification uses a donut of five minutes. Results for bandwidths of 30, 60, 120 and 180 minutes and donuts of 0, 5 and 10 minutes are in `empirics/results/rd.json`.

Rank design (`09_hep_sample.py`, `10_hep_pairs.py`, `11_rank_design.py`). All papers with primary category hep-th, hep-ph, hep-ex, hep-lat, gr-qc, nucl-th, nucl-ex, astro-ph.CO or astro-ph.HE, announced in their regular batch, 2017–2022, outside holidays (101,392 papers; 87,694 submitted at least 15 minutes after the batch opened). The time since the batch opened is measured from the previous weekday's 14:00 ET cutoff.

Overlap measures (`neardup.py`). Within each primary category, papers from July 2016 to December 2024 are represented by TF-IDF vectors of title and abstract: sublinear term frequencies, English stop words removed, terms in at least two and at most 30% of the category's papers. For a random sample of 3,000 papers from 2018 on, we compute the cosine similarity to the nearest earlier paper in the category within 24 months. Its percentiles define the category's thresholds τ_q. For each focal paper we keep every later paper in the category within 730 days whose similarity is at least τ_{50}. A pair is a *self* pair if the two author lists share a name key. Half of all pairs at τ_{50} are self pairs, and at τ_{95} the share is 87%, which is why the overlap outcomes use lower thresholds than the descriptive flag.

Citations and references. INSPIRE-HEP's public API gives citation counts excluding self-citations and curated reference lists, linked by record identifiers (`inspire_fetch.py`). Coverage of the high-energy-physics sample is above 99.9%. For other fields we use the Semantic Scholar batch API (`s2_fetch.py`), whose reference lists are often withheld for arXiv papers (41% coverage among later overlapping papers). The cutoff design therefore reports all-field citations for the subsample fetched by the time of writing.

Name keys (`names.py`). Names are transliterated to ASCII, TeX accents removed, and keyed as first name plus last name for submitters and as first initial plus last name for authors (full first names where the parsed author list has them, for the rotation analysis). Name keys are used instead of database author identifiers, which were reissued for many authors in 2025. Shared names produce collisions. The designs either difference them out (endorsement, LLM adoption) or remove them explicitly (submission cap: names whose submissions in a half-month span several archives).

Endorsement (`06_entry_policies.py`). A paper has a first-time submitter if the submitter's name key has not submitted since 2015. It has a submitter with no prior authorship if, in addition, the submitter's initial key appears in no earlier author list. Weekly cells by announcement week and group (Mathematics, other sections), September 2024 to October 2026. Each week of the 2025–26 academic year is differenced against the same week of 2024–25.

CS review policy (`06_entry_policies.py`). A title is review-type if it contains survey, review, overview, tutorial, position paper, "position:", primer, roadmap, perspective, state of the art or systematic literature. The dose-response uses the 29 computer-science categories with at least 300 papers in each Nov–Oct year from 2023–24 to 2025–26. A paper has a first-time author if any author with a full first name has a key never seen before.

Submission cap (`05_cap.py`). Days 1–15 of each month, papers announced in their regular batch. A submission is excess if it is the submitter's third or later in the half-month, and clean if all of the name's submissions in that half-month fall in one archive and the submitter is not a proceedings account.

LLM-style writing (`08_llm_authors.py`). Marker words: delve, showcase, underscore (all inflections), intricate, intricacies, meticulous, pivotal, realm, notably, commendable, garner, multifaceted, seamless, invaluable, groundbreaking, tapestry, elucidate. Their share of abstracts was 2.1% in 2019, 3.0% in 2022, 10.6% in 2024 and 4.4% in 2026. The decline in 2026 suggests that authors learned to avoid them, and it is one reason to classify adoption early.

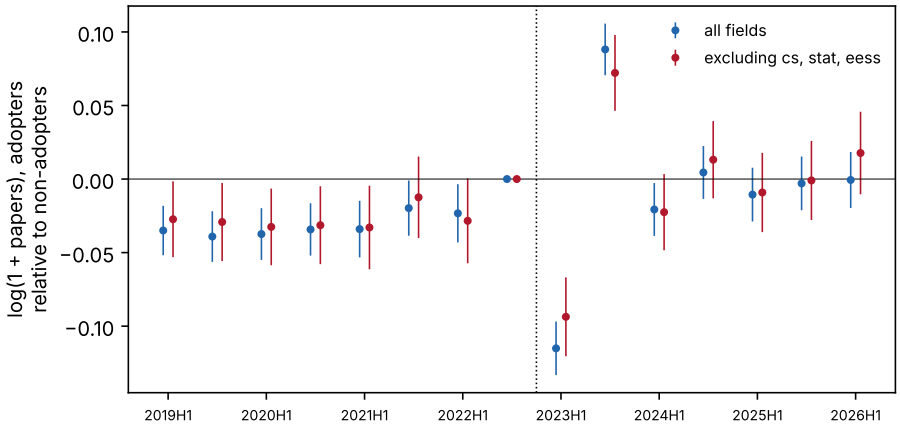


Figure 8: Author-level output of adopters of LLM-style writing, relative to non-adopters (event study, reference 2022H2; 95% intervals). Adoption is classified from 2023 abstracts. Blue: all fields; red: excluding cs, stat and eess.

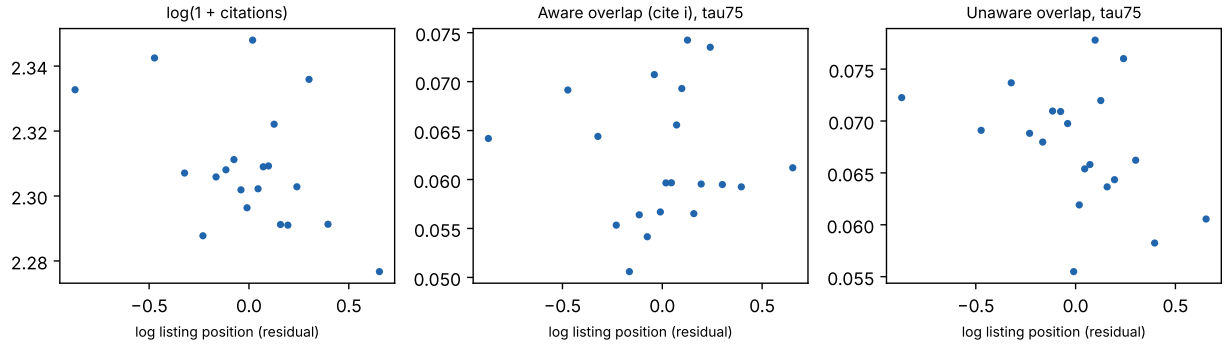


Figure 9: Listing rank in high-energy physics: binned residuals of outcomes on residual log listing position (twenty bins), net of category-year and submission-time fixed effects and log list size.